T3. NoSpherA2 — Understanding H-atoms in Crystallographic Refinement And Implementing Refinement Using Custom Aspherical Form Factors

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Outline

1. Setting the context
2. How H-atoms are treated in building the crystal structure model
3. HAR with NoSpherA2
4. Some Theory
5. Implementing NoSpherA2
6. More Tutorial Examples
7. Resources and Support

1.Setting the context

Setting the context

- **NoSpherA2 is aimed at making full use of good datasets, especially the best.**
- There are *many* purposes for crystallography in research.
- Different ‘quality markers’ are appropriate to these various purposes.
- I recommend reading Amber Thompson’s review for a perspective on the quality/benefit ratio of any sample that you need information on.

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Check for updates

Chemical Crystallography: when are ‘bad data’ ‘good data’?

Amber L. Thompson

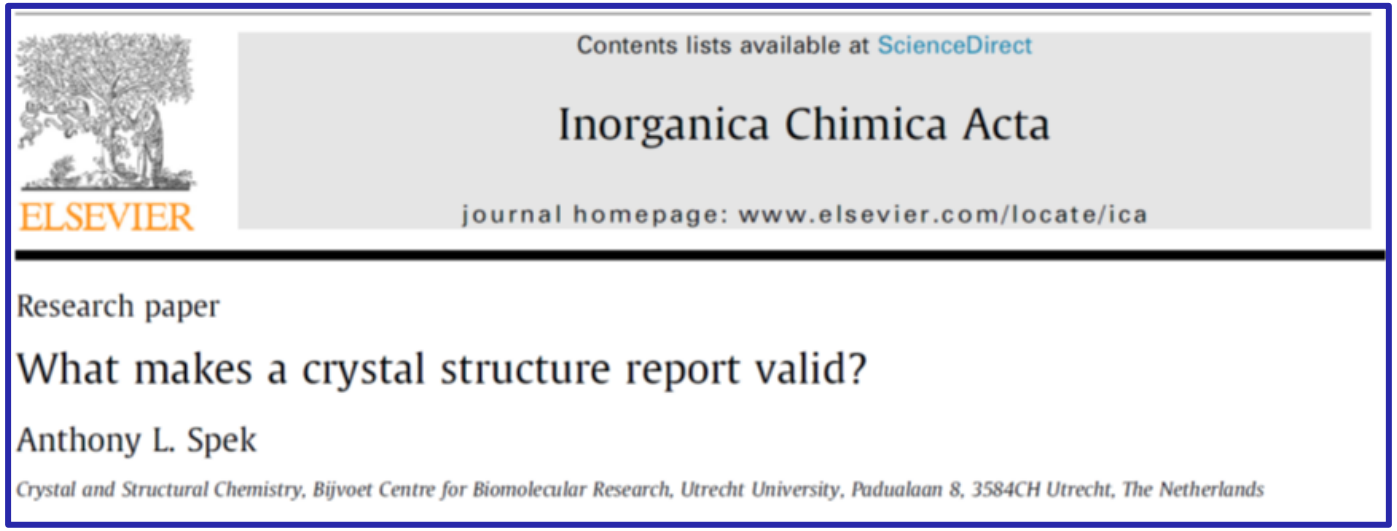
Chemical Crystallography, Chemistry Research Laboratory, Oxford, UK

The crystal quality of a surprising number of compounds can be improved, with practice and increasing skill. (Think back to Lecture 1 in this workshop.) The better the crystals, the better the data quality and the more useful that NoSpherA2 potentially becomes. Whether or not that effort is worthwhile is a judgement call that also comes with increased experience.

- Thus, many very weak crystals and ‘crystallographically flawed’ samples can be usefully investigated.
- Nevertheless, *many crystals* are produced with excellent crystallinity, and which diffract very well.
- Such datasets contain much more information than the standard model for refinement is able to use.
- NoSpherA2 is one technique for extracting some of this information.

Setting the context (2)

- Make use of *all* help – including CheckCIF!
- Different ‘quality markers’ are appropriate to these various purposes.
- Understand the *purpose* of the ‘validation report’ – e.g. read Ton’s paper: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Research paper

What makes a crystal structure report valid?

Anthony L. Spek

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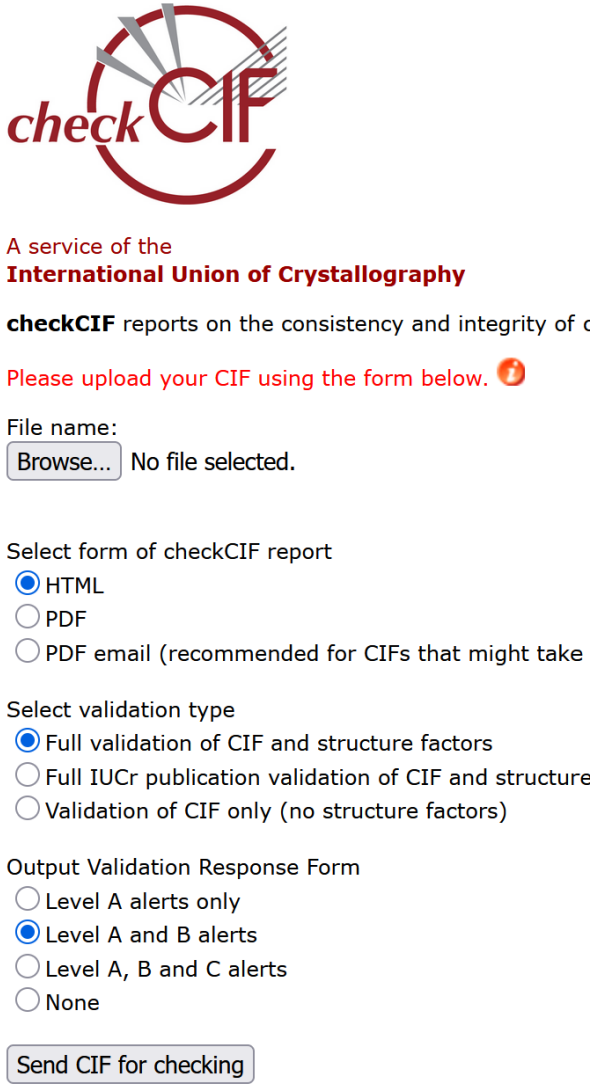
- https://checkcif.iucr.org/ (for manual submission) or automatically from Olex2)
- HTML option is helpful to get links for the meaning of the Alerts.
- Try to fix all level ‘A’ and ‘B’ alerts.
- If they cannot be fixed, use the Validation Response Form



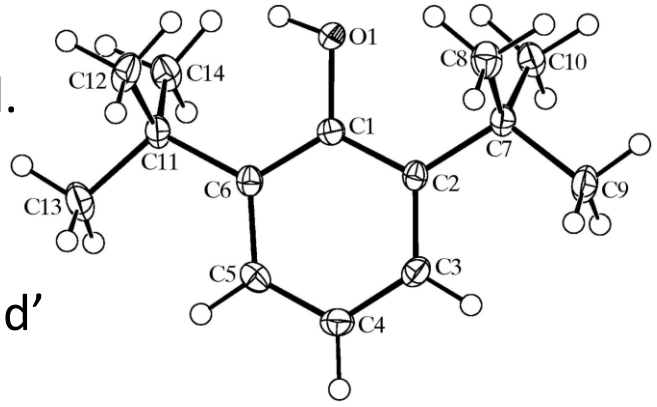
Commit to Best Practices and Advanced Understanding

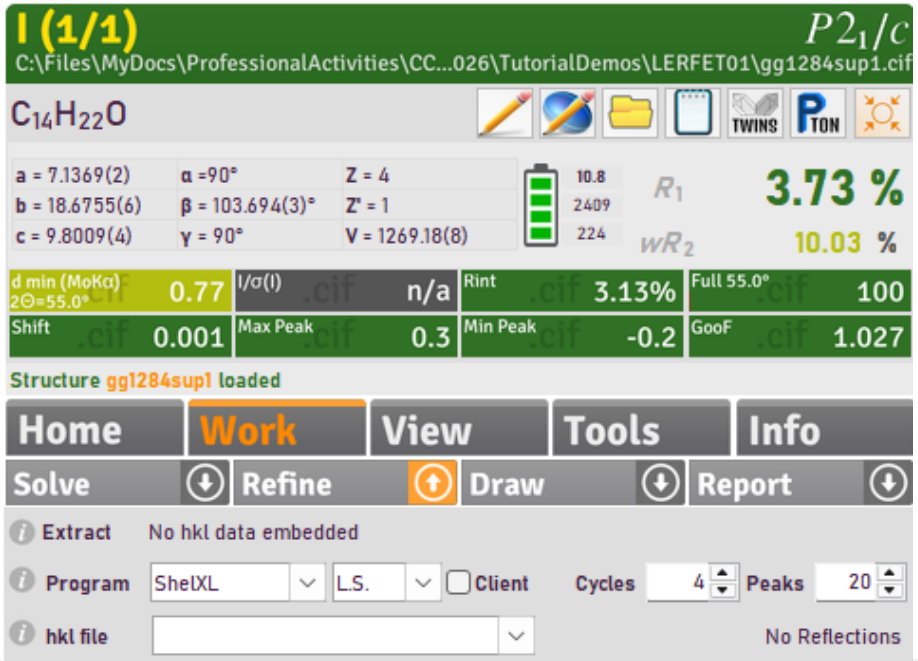
- We recommend OLEX2 because it is a multifaceted Graphical User Interface (GUI) for SCXRD
- Olex2 is designed to be *easy to use for beginners* and highly recommended to you by the CCCW
- Yet it is a highly sophisticated software package and far more than just a GUI
- Therefore, *as you mature in SCXRD*, commit to learning more about what Olex2 does
- Use the www.olexsys.org support pages
- Make use of the excellent third-party user manual maintained by Dr. Ilia Guzei of the University of Wisconsin – Madison campus
  https://uwmadison.app.box.com/s/2qq12zzw893vc3n1rspefv9h9p6gw83k
- Ilia gives hundreds of tips and will help you go become a power user
- Just the software installation tips on pages 4-16 of this manual are totally worth the effort of downloading
- I strongly advocate installing the Zürich Space Group Helper and using it for *every* dataset that you process

2. How H-atoms are treated in building the crystal structure model

What does ‘Add Hydrogens’ do?

- We take a real example: 2,6-di-tert-butylphenol, a simple organic compound.
- It was first reported in 1992 with CSD ‘refcode’: LERFET
- In 2005, Lutz and Spek redetermined the structure because it, in their words, ‘features an OH moiety that does not appear to form a hydrogen bond’
- The new version is called LERFET01, for which intensity data is available at the journal website, https://doi.org/10.1107/S0108270105030325
- The files we can download are: gg1284sup1 and gg1284Isup2.hkl
- Let us open this structure in Olex2 from the CIF file: it opens to reveal a good ‘R1’ value of 3.73%.
- More importantly, it has great data quality (Rint = 3.13%; complete data to 2θ = 55.0°, v. small final diff. peak/hole)
- However, we must *associate* the hkl file to the structure.





LERFET: G.G. Lazarev e.a., *Z.Phys.Chem.NeueFolge*, 1992, *176*, 41.

LERFET01: M. Lutz, A.L. Spek, *ActaCrystC*, 2005, *61*, *o639*.

LERFET demo (1)

- We start by *re-solving* the structure; then name the atoms and sort the list.
- Move to refine with SHELXL, 4 cycles, and peaks = 40 (~2x the expected number of H atoms)
- SHELXL implements a refinement in the **spherical atom approximation** and uses **neutral atomic scattering factors**

❖ All atoms are treated as if all their electrons are distributed about the nucleus, i.e. a gas phase isolated atom structure, and #*e* = *Z* (the atomic number of the element)

❖ All atoms are treated as if neutral (i.e. ignoring ionic charges), a non-necessary simplification adopted for overall convenience in molecular crystallography.

- It was written by genius crystallographer George Sheldrick†

In Memoriam: George M. Sheldrick (1942–2025†)

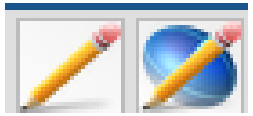
It is with profound sadness that the scientific community announces the passing of George M. Sheldrick, a pioneering crystallographer whose innovative contributions have left an indelible mark on the world of science. Born in 1942, George dedicated his career to advancing the field of crystallography, transforming the way scientists determine and understand molecular structures.

LERFET demo (2)

- Next, we improve the non-H atom refinement by allowing **anisotropic displacements** about the c.g.
- Now consider the Sheldrick code for this structure (the 'RES file')

```
FVAR 0.80843
O1    O     0.594159  0.430222  0.463342  11.00000  0.01677  0.02212 =
 0.01420 -0.00382  0.00174  0.00698
C1    C     0.751793  0.385698  0.469750  11.00000  0.01331  0.01263 =
 0.01298  0.00131  0.00454  0.00007
.
.
C14   C     0.719981  0.403277  0.783979  11.00000  0.03701  0.02581 =
 0.01351 -0.00228  0.01245 -0.00245
```

Omitting the next 12 C atoms in identical format…

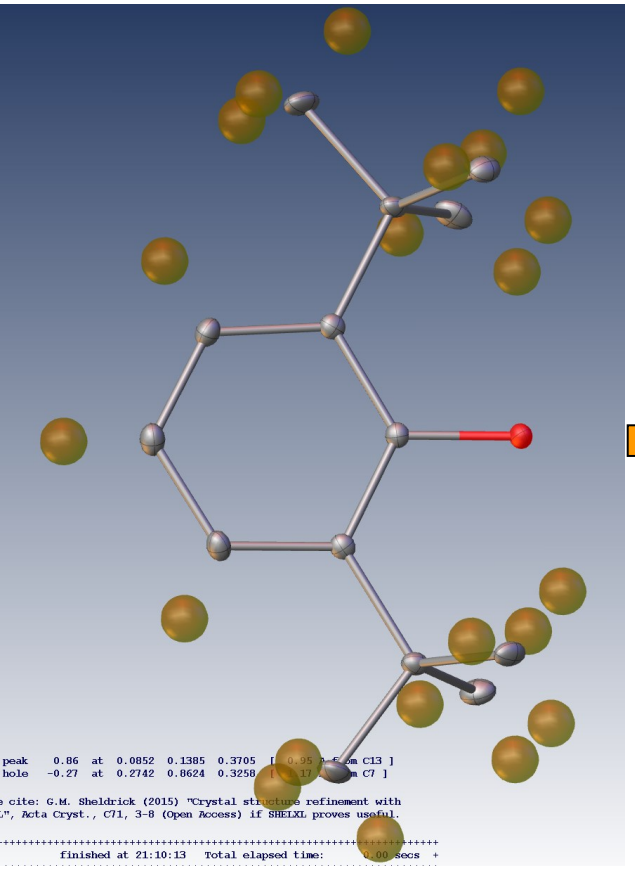


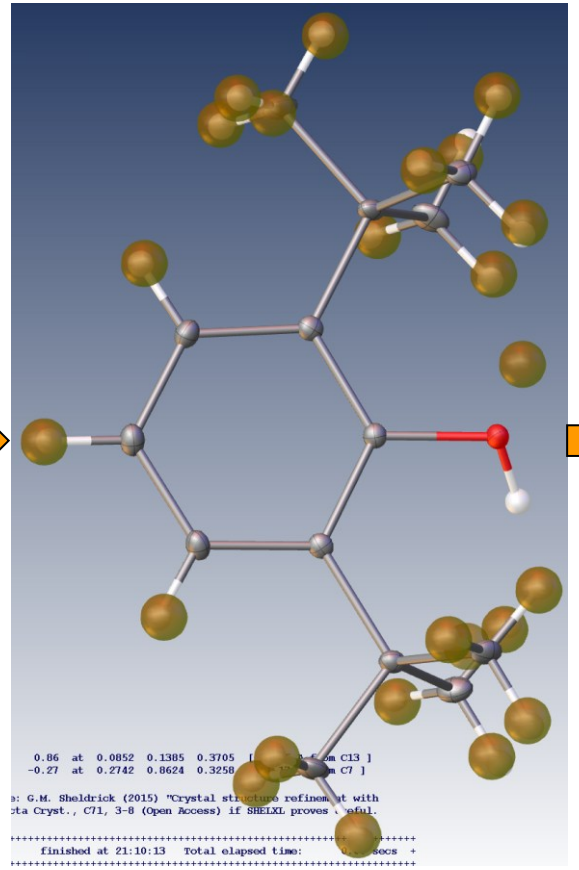
Olex2 'windows' into the Shelx Instructions and Results parameters. Start looking at this for insight into how Olex2 organizes and reports the structures.

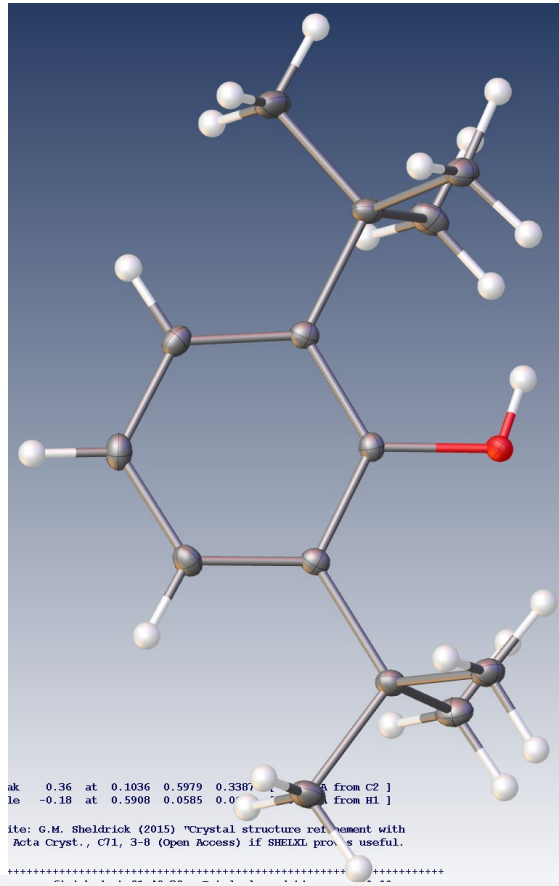
- FVAR 'scales' the experimental scattered X-ray intensity to the No. *e* in the structure (called F000).
- O1 and C1-C14 are *arbitrary* labels up to 4-characters; O and C, *required*, define the element types
- 0.594159 0.430222 0.463342 are the atom *fractional coordinates* in the unit cell.
- 11.00000 is shorthand, the first "1" is the VAR (variable) identity, then "1.00000" defines the *occupancy*
- 0.01677 0.02212 0.01420 -0.00382 0.00174 0.00698 are the 6 tensor components for the 'football' shapes of the anisotropic displacement refinement

LERFET demo (3)

- Now we scroll the mouse wheel to reduce the number of peaks (removing the *lowest* intensity peaks)
- We easily get to *exactly* 22 brown spheres (peaks) that represent all the H atoms (scroll over for intensities: 0.86 to 0.69 $e/Å^3$, which is *remarkable, excellent diffraction data quality*)
- Type: HADD <retn> to see the H atoms ‘fit’ the spheres; then Refine to absorb the ca. 17 *e* detected.







LERFET demo (4)

- Look again at the Sheldrick code for this stage of the model: (the 'RES file')

```
FVAR 0.86527
O1    O     0.593857  0.430053  0.462770  11.00000  0.02013  0.02739 =
 0.01557 -0.00619  0.00167  0.00726
AFIX 147
H1    H     0.598482  0.447466  0.542523  11.00000 -1.50000
AFIX 0
.
C3    C     1.058082  0.337708  0.589645  11.00000  0.01869  0.01999 =
 0.01894  0.00406  0.00138  0.00127
AFIX 43
H3    H     1.161978  0.334319  0.670411  11.00000 -1.20000
AFIX 0
.
C8    C     0.565027  0.426404  0.155221  11.00000  0.02676  0.02135 =
 0.01585  0.00083  0.00067 -0.00047
AFIX 137
H8A   H     0.539646  0.459297  0.226656  11.00000 -1.50000
H8B   H     0.457398  0.428133  0.071944  11.00000 -1.50000
H8C   H     0.684253  0.440614  0.129406  11.00000 -1.50000
AFIX 0
```

147 instruction fits OH *and* forces it to adjust to available ED

-1.5 instruction assigns *displacement* of the H to 1.5X that of O1

C3 is the first aryl C; 43 instruction makes a planar H for sp^2 C

-1.2 instruction assigns *displacement* of the H to 1.2X that of C3

C8 is the first methyl C; 137 instruction rotates sp^3 CH_3 to fit the ED

-1.5 instruction assigns *displacement* of each H to 1.5X that of C8

We can inspect the structure to see the imposed model:

d(O-H) = 0.840 Å

Uiso = 0.032, vs 0.021 for O1

Neutron *d*(O-H) = 0.989 Å

d(C-H) = 0.950 Å

Uiso = 0.024, vs 0.020 for C3

Neutron *d*(C-H) = 1.085 Å

d(C-H) = 0.980 Å

Uiso = 0.033, vs 0.022 for C8

Neutron *d*(O-H) = 1.084 Å

LERFET demo (5)

- The R-value we now get is 4.00%, but Lutz&Spek *refined* H atoms!
- Type: sel $H, then AFIX 0 <retn>; then refine 30 cycles
- Type: sel $H, then FREE Uiso <retn>; refine 30 cycles; close!
- Now review the .RES coding for the structure:


```
FVAR 0.86792
O1    O     0.593817  0.430086  0.462957  11.00000  0.01981  0.02686 =
 0.01634 -0.00580  0.00164  0.00680
H1    H     0.609969  0.450904  0.543397  11.00000  0.05811
.
C3    C     1.057967  0.337717  0.589661  11.00000  0.01909  0.02030 =
 0.01903  0.00386  0.00180  0.00093
H3    H     1.167987  0.334052  0.670339  11.00000  0.02942
.
C8    C     0.565099  0.426459  0.155307  11.00000  0.02643  0.02161 =
 0.01616  0.00143  0.00056 -0.00026
H8A   H     0.536141  0.462722  0.228288  11.00000  0.03258
H8B   H     0.458406  0.428260  0.067090  11.00000  0.03296
H8C   H     0.682713  0.439984  0.126973  11.00000  0.02651
```

d(O-H) = 0.862 Å
Uiso = 0.058, vs 0.021 for O1
Neutron *d*(O-H) = 0.989 Å
AFIX 147/AFIX 0 *gone*

d(C-H) = 0.976 Å
Uiso = 0.029, vs 0.020 for C3
Neutron *d*(C-H) = 1.085 Å
AFIX 43/AFIX 0 *gone*

d(C-H) = 1.041 Å
Uiso = 0.033, vs 0.022 for C8
Neutron *d*(C-H) = 1.084 Å
AFIX 137/AFIX 0 *gone*

Bond precision: C-C = 0.0015 A

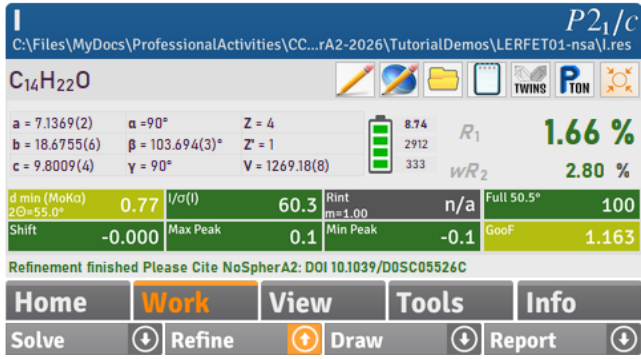
Summary: How H-atoms are treated in the IAM

- The spherical atom model, usually called "independent atom model" or IAM, fails for H *because its electron density (ED) distribution is never near-spherical* as it has zero 'core' electrons.
- ❖ Earliest structures: H atoms were just ignored (often mineral or metal complex structures); but in 2,6-di-tert-butylphenol, the 22 H atoms are 22/114, > 19% of the total *e* count
- ❖ Better data quality: treat H atoms using the *Riding Model*, which the HFIX commands implement.
- ❖ Best data quality: freely refine *either* or *both* H-atom XYZ coordinates and H-atom isotropic displacements, but unable to locate the H nuclei accurately (cf. neutrons…)
- The IAM (using *spherical, gas-phase, isolated ('independent') atoms to compute scattering factors from 1950's era Hartree-Fock-Slater quantum calculations*) was developed when the raw data quality was the actual 'limiting factor' for structures determined by X-ray diffraction.
- *Neutron diffraction* was able to determine accurate H-atom positions *and* displacements, but experiments were difficult, very expensive, and demanded HUGE crystals (10X larger by volume)
- Today, many if not most X-ray diffraction data sets *contain much more information than the IAM model is able to make use of.* The *model* is now limiting the accuracy of the experiments.

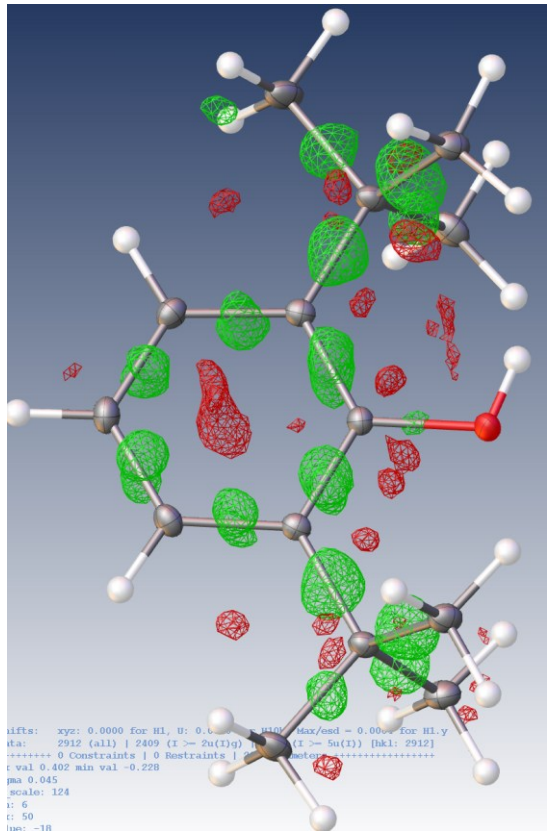
N.D.D. Hill and R.T. Boeré, *Chemistry–Methods*, 2025, *5*, e202400052.

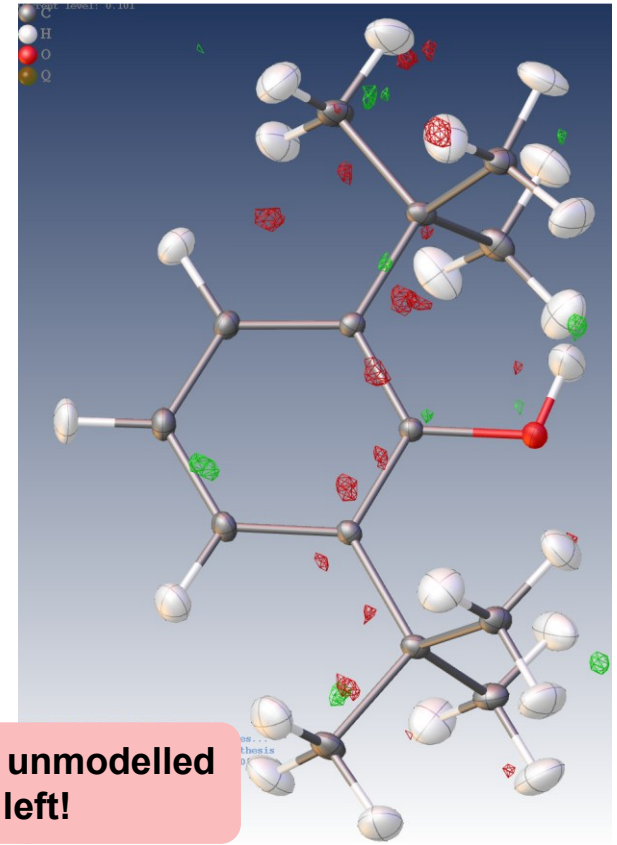
3. HAR with NoSpherA2

HAR with NoSpherA2



- Now let us see what quantum crystallography can do!
- The data quality is evidently magnificent, so we have high hopes.
- Switch to olex2.refine and fully optimize; inspect ‘CRTL+M’ map *residual ED*.
- Lots of unmodelled ED in ‘final refinement’ IAM model, e.g. the C-C and C=C bonding ED (green)





***NO* unmodelled ED left!**

d(O-H) = **0.934Å**
Uiso = 0.049, vs 0.020 for O1
Neutron *d*(O-H) = 0.989 Å

d(C-H) = **1.093 Å**
Uiso = 0.035, vs 0.018 for C3
Neutron *d*(C-H) = 1.085 Å

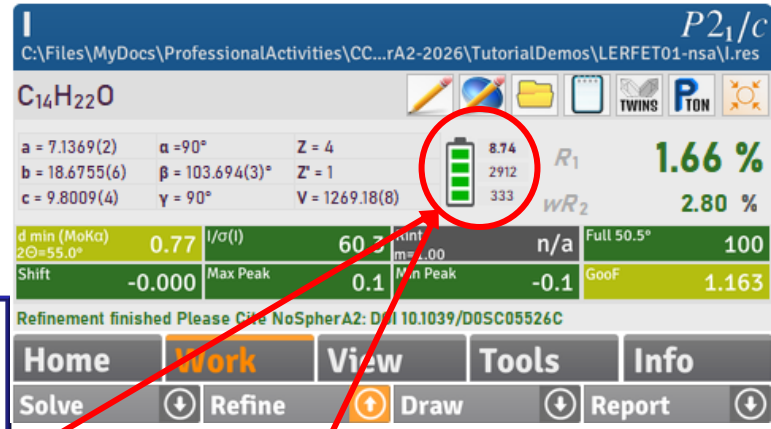
d(C-H) = **1.091 Å**
Uiso = 0.042, vs 0.021 for C8
Neutron *d*(C-H) = 1.084 Å

Bond precision: C-C = 0.0006 Å, 60% improv. over the IAM model, i.e. *effects beyond H*

LERFET Final Result

- Consider the final .RES coding for the structure:

```
FVAR 0.845799
O1    O     0.594336  0.429973  0.463412  11.00000  0.01891  0.02469 =
 0.01496 -0.00457  0.00134  0.00582
H1    H     0.607837  0.454681  0.548056  11.00000  0.04691  0.05899 =
 0.03578 -0.01851 -0.00294  0.01998
.
C3    C     1.058220  0.337660  0.590058  11.00000  0.01739  0.01943 =
 0.01676  0.00230  0.00177  0.00334
H3    H     1.179309  0.333111  0.681875  11.00000  0.02729  0.04300 =
 0.03159  0.00095 -0.00118  0.00501
.
C8    C     0.564849  0.426460  0.155303  11.00000  0.02587  0.01939 =
 0.01536  0.00156  0.00047  0.00035
H8a   H     0.538788  0.465341  0.231829  11.00000  0.06081  0.02684 =
 0.03739 -0.00502  0.00915  0.00982
H8b   H     0.447078  0.428923  0.061920  11.00000  0.04784  0.04286 =
 0.03610  0.00271 -0.01310  0.00060
H8c   H     0.695878  0.442663  0.125390  11.00000  0.04009  0.04123 =
 0.04720  0.00703  0.01220 -0.00783
```



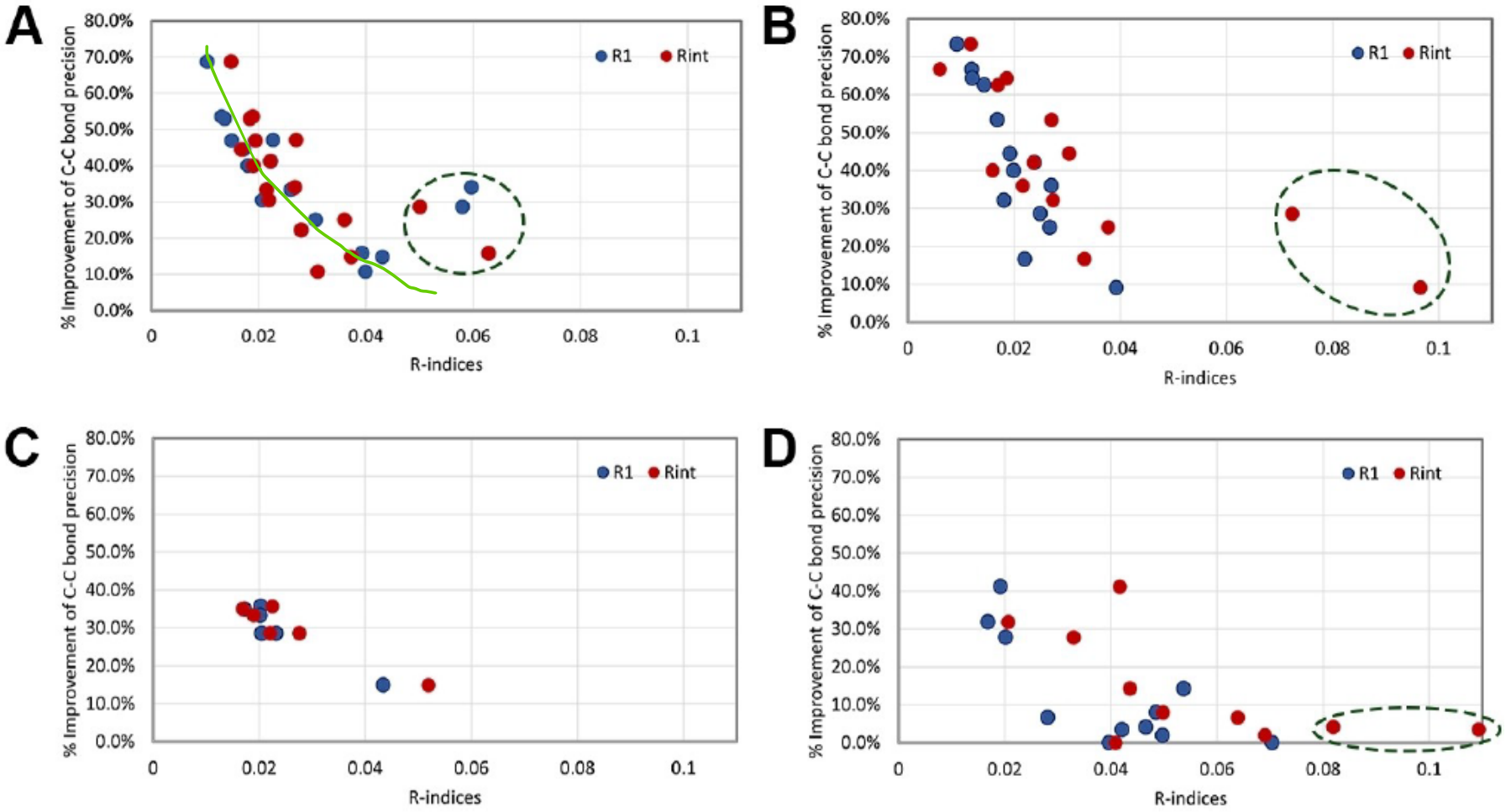
Data:param. “battery”: need 8:1

For LERFET, we were able to refine *all* the H-atoms with anisotropic displacement parameters. The file structure is now homogeneous with 9-parameters for each atom in the structure.

Note that most of the *improved precision* can be obtained even if the H-atoms must be left isotropic.

Beyond the H-atoms: Improved Precision with Aspherical Form Factors

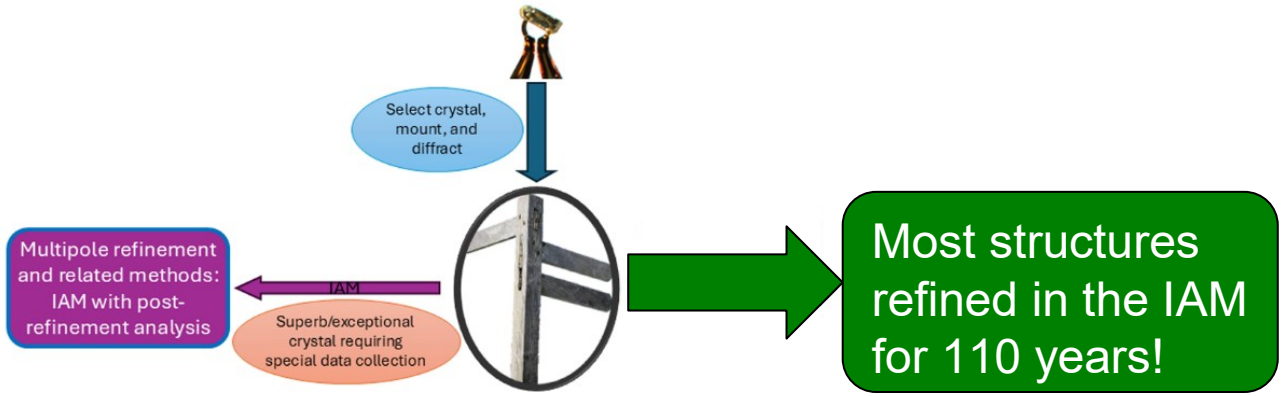
Correlation diagrams for the improvement in refinement precision and R-factors



N.D.D. Hill and R.T. Boeré, *Chemistry–Methods*, 2025, *5*, e202400052.

- **Ave C-C precision: 60% improv.**
- The *standard uncertainty of L.S. refinement is the most important measure of model quality in chemical crystallography.*
- Why are the C–C bonds improv.?
- Not all C atoms are identical!
- Single, double, triple and aromatic bonds
- *Partial charges*! Assumption of all atoms being neutral is wrong.
- Cumulative effect of assumptions degrades model quality.
- Calculating custom aspherical form factors for *each unique atom* in a structure has spin-off benefits
- Expected effect is greatest with lowest atomic numbers…

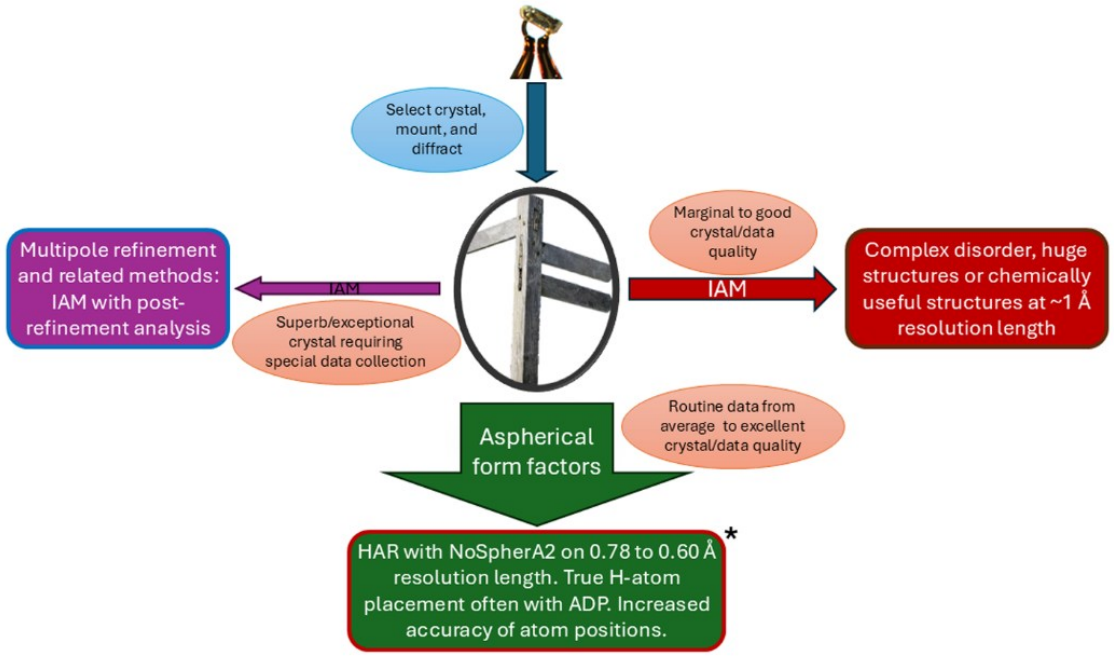
Alternatives to the IAM in X-ray crystallography



- Limitations of the IAM in crystallographic refinement known for a *long* time.
- Proposed solutions: McWeeny [McWeeny, R. (1952). Acta Cryst. 5, 463-468; (1953). Acta Cryst. 6, 631-637.]; Dawson [Dawson, B. (1967). Proc. R. Soc. London Ser. A, 298, 255-263.]; Kurki-Suonio [Kurki-Suonio, K. (1968). Acta Cryst. A24, 379-390.]; Hirshfeld [Hirshfeld, F. L. (1971). Acta Cryst. B27, 769-781.; (1977). Theor. Chim. Acta, 44, 129-138.]; Stewart [Stewart, R. F. (1976). Acta Cryst. A32, 565-574.]; Hansen & Coppens [Hansen, N. K. & Coppens, P. (1978). Acta Cryst. A34, 909-921.] – Which lead to the Hansen-Coppens pseudo-atom model, which works best with *multipole refinement*.
- Since *Multipole Refinement* allows for deformation of electron density, this is a good solution.
- In practice, however, multipole refinement is extremely demanding, requiring exceptional crystal quality, long data acquisition and complex refinement procedures.
- It remains a crucial niche technique used especially to model the *valence electron densities, i.e. studying bonding*

W.F. Sanjuan-Szklarz *et al.*, *IUCrJ*, 2020, *7*, 920.

The “crossroads” in small molecule X-ray crystallography



- ❑ Introduction of new ‘quantum crystallography’ methods and software makes using ‘routine’ aspherical form factors feasible
- ❑ Majority of modern XRD datasets contain more information than “independent atom model” (IAM) can harness.
- ❑ Provides *excellent motivation* to pursue the best quality crystals in the lab and obtain best-quality XRD datasets always.
- ❑ HAR using NoSpherA2 in Olex2 now offers a good alternative to the IAM for *most* XRD datasets (cf. Amber Thompson!)
- ❑ We have written a guide for ‘users’ – i.e. practical introduction for chemical crystallographers – to the NoSpherA2 method.

N.D.D. Hill and R.T. Boeré, *Chemistry–Methods*, 2025, *5*, e202400052.

Summary: Reasons to Adopt Aspherical Form Factor Refinements

Standard X-ray refinement involves data fitting to *spherical atom model* (IAM)

- Assumes that the atoms are isotropic and that the atomic positions are the maxima in the ED.
- Details of covalent interactions are missing in the model. The X-ray data carries this information. This is especially true for hydrogen atoms, which lack inner cores of electrons and have a single valence electron.

Aspherical Form Factor Refinements involves quantum-mechanical determination of electron density from wave function-based calculations. The electron density is the square of the absolute value of the wave function. The interpretation of X-ray data is then carried out in terms of Hirshfeld scattering factors from the Quantum-mechanical model.

- It is, above all, *not* a 'computational structure determination'! We only compute the atomic scattering factors.
- Drastically improves the refined models (e.g. R1 = 3% for IAM, 0.7% for NoSpherA2).
- For accurate charge densities it allows separating harmonic and anharmonic thermal motion effects from density deformations due to bonding.
- Precise location and refinement of hydrogen atoms without neutron radiation.
- Allows refining positions of hydrogen atoms bound to transition or heavy elements.
- Often featureless residual density (i.e. all the information that is in the X-ray data has been extracted).
- The *weights* ('fudge factors') used in refinement become negligible as the model is now so good.

N.D.D. Hill and R.T. Boeré, *Chemistry–Methods*, 2025, *5*, e202400052.

4. Some Theory

The original paper that introduced NoSpherA2

Chemical Science

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Accurate crystal structures and chemical properties from NoSpherA2†

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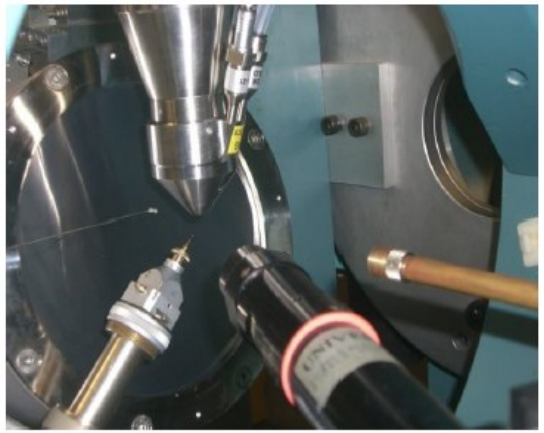
This paper is worth reading through carefully. It is presented in an approachable, largely non-mathematical way and treats a number of interesting examples that illustrate a wide range of applications.

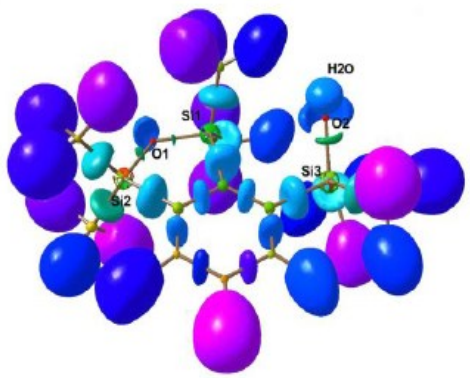
- ❑ Detailed 18-page article, with historical introduction to the state of the field in 2020.
- ❑ Thorough explanation of the theory for how NoSpherA2 works.
- ❑ Seven pages of applications that illustrate the ability of NoSpherA2 in very different kinds of substances: organic crystals; boron hydrides, CaF2, osmium hydride.
- ❑ Ability to refine H-atom xyz coordinates reliably – to ‘neutron’ bond lengths.
- ❑ Possibility of anisotropic displacement ellipsoids for H-atoms (impossible in the IAM).
- ❑ Correct treatment of non-bonded ED: lone pairs, metal *d* and *f* electrons

NoSpherA2 is a product of *quantum crystallography*

- Recognition that XRD *experimentally* measures electron density (ED)
- Quantum mechanics *computes* ED by the square of the wavefunction



+ $|\Psi(r,t)|^2$ =



Measure: $I(hkl) \propto F_{obs}^2(hkl)$

Model from: $F_{obs}(hkl)\ vs.\ F_{calc}(hkl)$

110 years of X-ray diffraction by crystals

$$R = \frac{\sum \left[|F_{obs}| - |F_{calc}| \right]}{\sum |F_{obs}|}$$

S. Grabowski, et al., Chem. Sci. **2017**, *8*, 4159.

F. Kleemiß, et al., Chem. Sci. **2021**, *12*, 1675.

So what is this F_{calc} term?

- Structure factor equation (a phase rule):

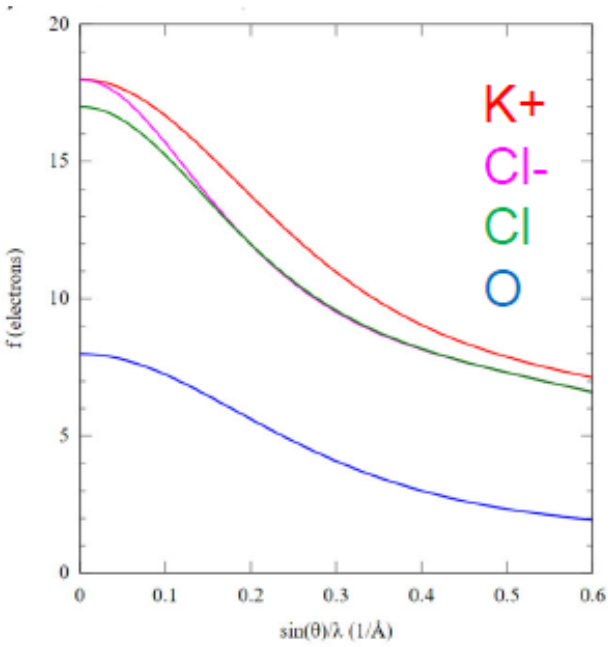
$$F_{hkl} = \sum_{j=1}^{N} f_j e^{[2\pi i(hx_j + ky_j + lz_j)]}$$

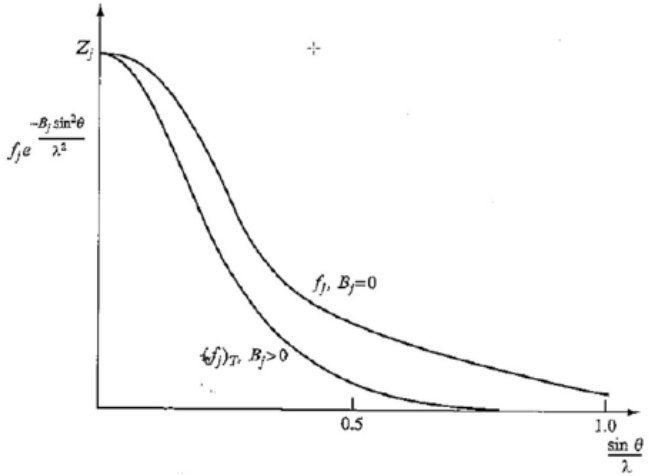
- Atomic scattering (form) factor f^o for each atom *j:*

$$f^0 = \frac{\lambda}{\sin\theta}\int_0^{\infty} \rho(r) r \sin\left(\frac{4\pi\sin\theta}{\lambda}\right)^2 dr$$

- Modified by the Debye Waller factor for motion of the atom (vibrations) that further decreases intensity with sinθ/λ

$$F_{hkl} = \sum_{j=1}^{N} f_j e^{[-B_j\,(2sin\theta/\lambda)^2]} e^{[2\pi i(hx_j + ky_j + lz_j)]}$$





Refer to L5 and L7, on Structure Factors, by Joe Ferrara!

Where do atomic scattering (form) factors come from?

- At first, form factors were determined empirically, from response of standard samples for decay of X-ray scattered intensities with (sin θ)/λ
- Later replaced by self-consistent field (SCF) computation: early versions of computational chemistry based on 1950's era quantum mechanics

International Tables for Crystallography

Intensity of diffracted intensities

P. J. Brown, A. G. Fox, E. N. Maslen, M. A. O'Keefe and B. T. M. Willis. *International Tables for Crystallography* (2006). Vol. C, ch. 6.1, pp. 554-595

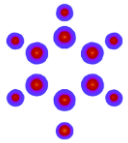
[doi:10.1107/97809553602060000600]

Abstract

Section 6.1.1 covers X-ray scattering from atoms and ions. Scattering is described by the Thomson formula, including coherent (Rayleigh) and incoherent (Compton) X-ray scattering. Atomic scattering factors, calculated using relativistic Hartree–Fock or Dirac–Slater wavefunctions, give the X-ray scattering from

"calculated using relativistic Hartree-Fock or Dirac-Slater wavefunctions …"

- Modern IAM-based Least Squares refinement programs for X-ray crystallography use tabulations of f_j from such sources to model each *element* type only, irrespective of where it is in the structure



IAM = Independent Atom Model

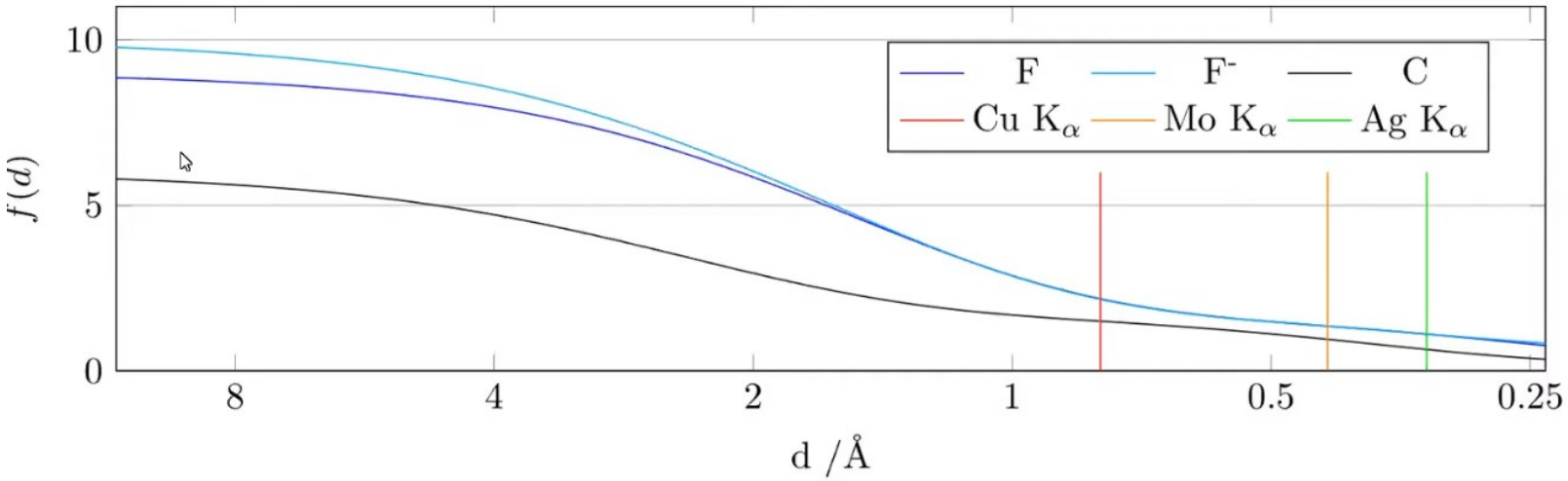
- IAM is *useful* (reasonably accurate and very effective) but it *is an approximation:*
  1. Treats each atom of a given element identically (and as a neutral atom)
  2. Ignores subtle (and gross) differences in atomic charge distributions in space
  3. (Must) treat hydrogen as if the nucleus is at the centre of the polarized ED

The IAM is to crystallography what the Ideal Gas Law is to thermodynamics: a simple, very robust and easy to implement starting model…

Scattering factors w.r.t. resolution

We have fabulous hardware:

- SuperNova dual Cu/Mo with Pilatus 200K
- This is 100x better than the instruments of 1960's
- Yet the *models* we build depend on the *same* spherical atom approximations for the form factors as used with hardware built in the 1960s!
- Why is it important to go beyond the IAM model?
- The 'atomic form factor' f_j : neutral F atom is not identical with the F^- form factors
- A significant difference *for the low-resolution data!*

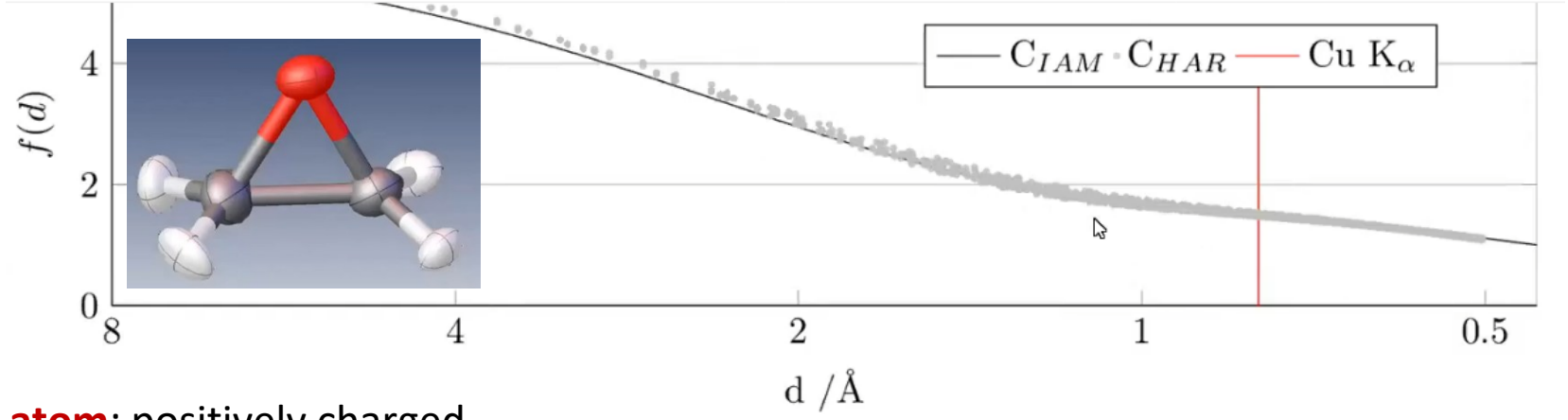


F. Kleemiß, et al., Chem. Sci. **2021**, *12*, 1675.

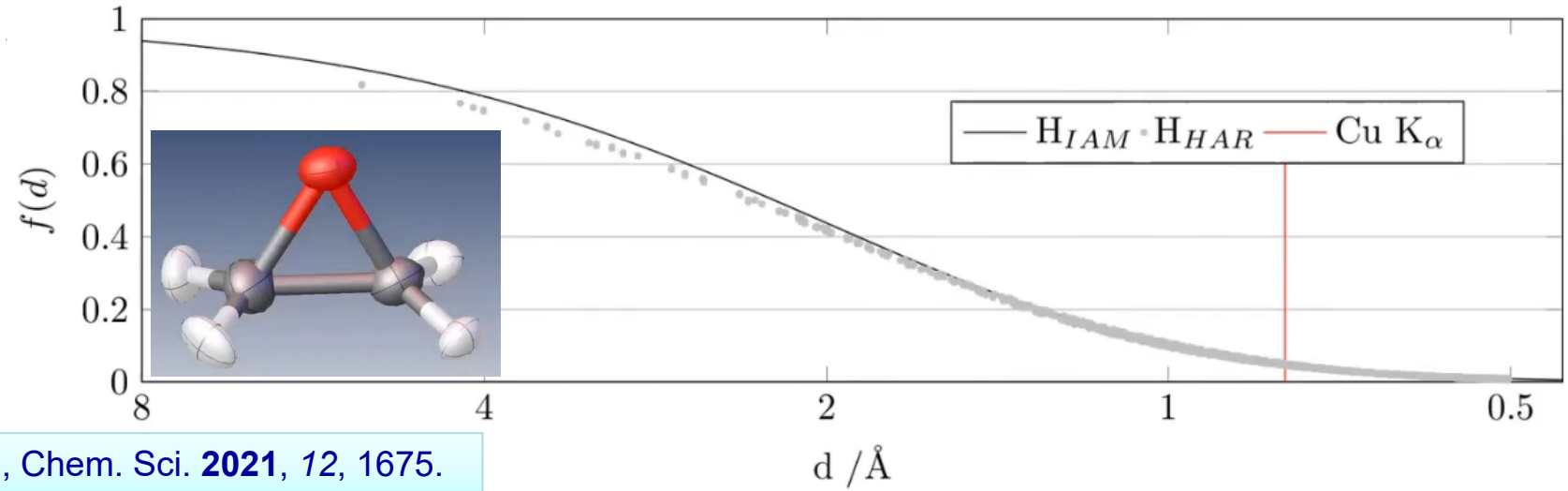
Aspherical form factors also important for neutral atoms

- Scattering factor curves for the IAM model (solid line) versus the corrected asphericity

Carbon atom: negatively charged



Hydrogen atom: positively charged

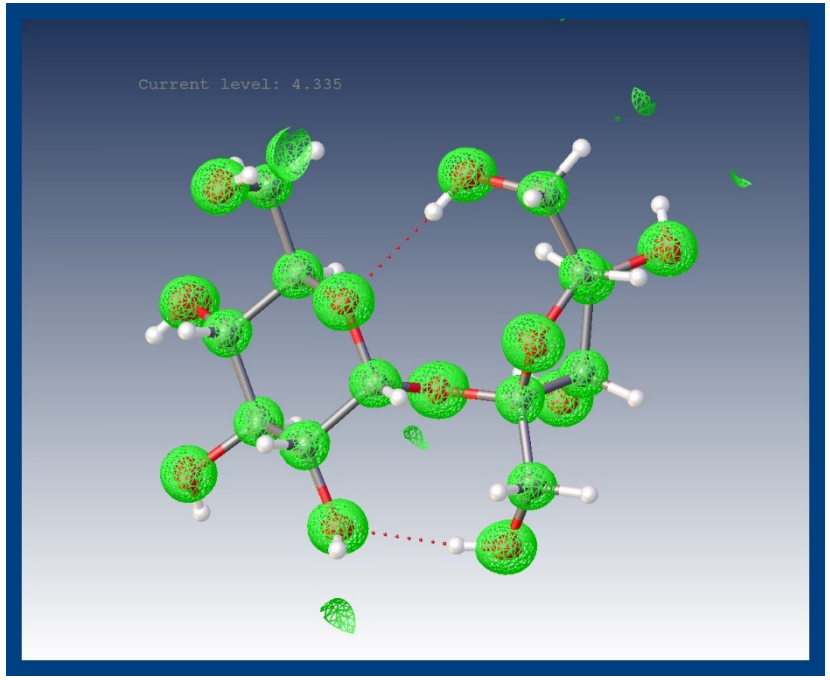


Organic chemists pay inordinate attention to the *partial atomic charges* of the atoms in their molecules.
These also affect the atomic scattering factors in XRD.
Partial charges are accurately computed by DFT methods.

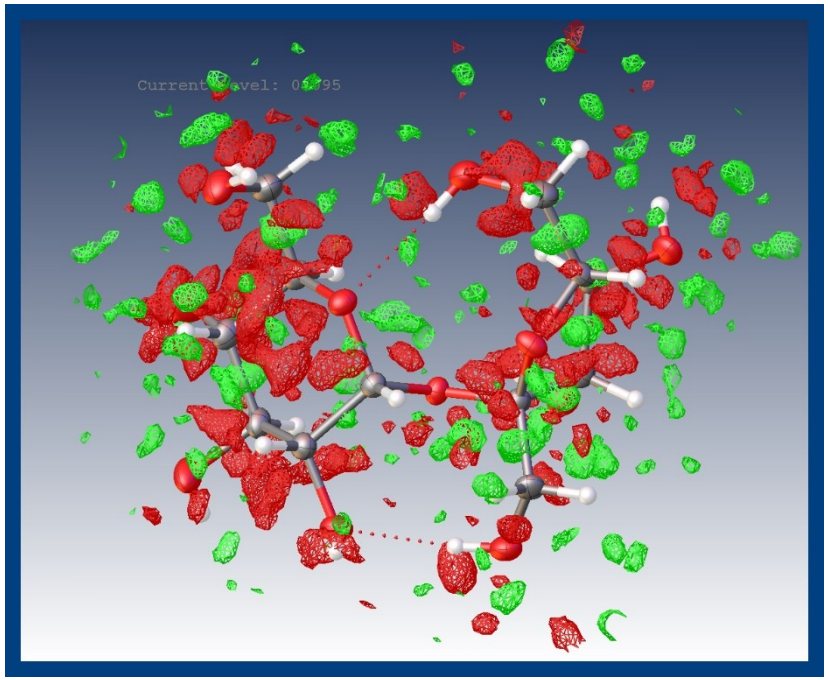
F. Kleemiß, et al., Chem. Sci. **2021**, *12*, 1675.

Consequences of the IAM approximation

1. Residual electron density:



Sucrose: a map of *F*(calc) = assigned ED



Sucrose: final difference map of [*F*(obs) – *F*(calc)] = unassigned ED

2. *Weighting factors* required in refinement
3. *R*-factors that cannot go to zero!
   A 'good structure' is one where $R_1 < 5\%$
4. If the data quality is good, we are losing structural information.
5. Atoms defined by low electron counts are poorly resolved – hydrogens!

What does ‘HAR’ mean?

- Hirshfeld Atom Refinement is the *method of charge partitioning* which Hirshfeld proposed and named as the ‘stockholder partitioning scheme’

Theoret. Chim. Acta (Berl.) 44, 129–138 (1977)

THEORETICA CHIMICA ACTA
© by Springer-Verlag 1977

Bonded-Atom Fragments for Describing Molecular Charge Densities

F. L. Hirshfeld

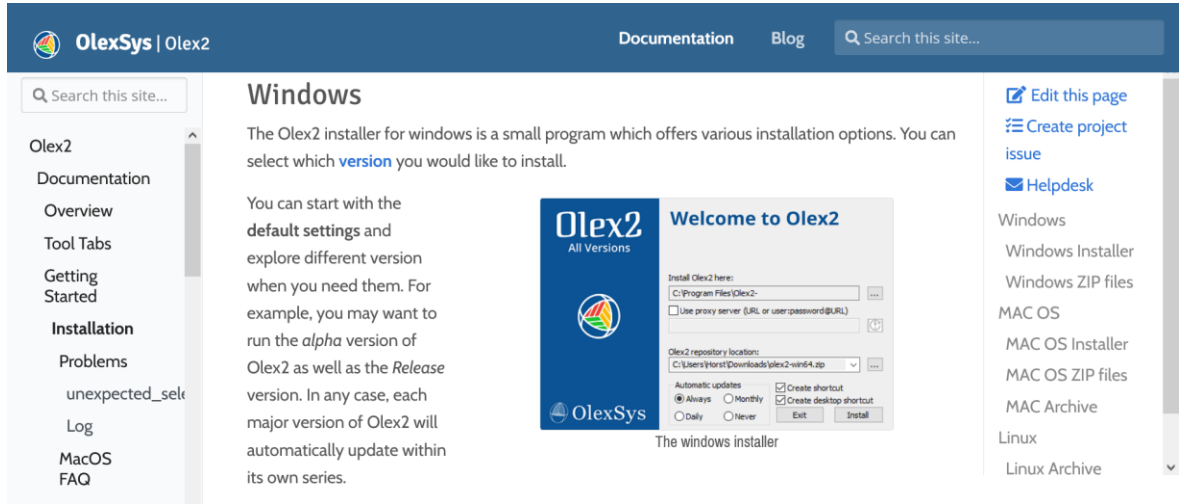
Department of Structural Chemistry, Weizmann Institute of Science, Rehovoth, Israel

- HAR is only a *method* for calculating the atomic scattering factors from the DFT wavefunction computation and is not essential to the concept.
- Florian Kleemiss is working on new “Grid-free quantum crystallographic refinement using unbiased analytic atomic form factors from density fitting” in a paper that will be published soon.
- The goal: “By avoiding real-space grids and numerical integration, the method substantially reduces computational cost while retaining a physically meaningful description of atomic electron densities.”
- This should allow the treatment of substantially larger molecules with custom aspherical form factors.

5. Implementing NoSpherA2

Getting Started

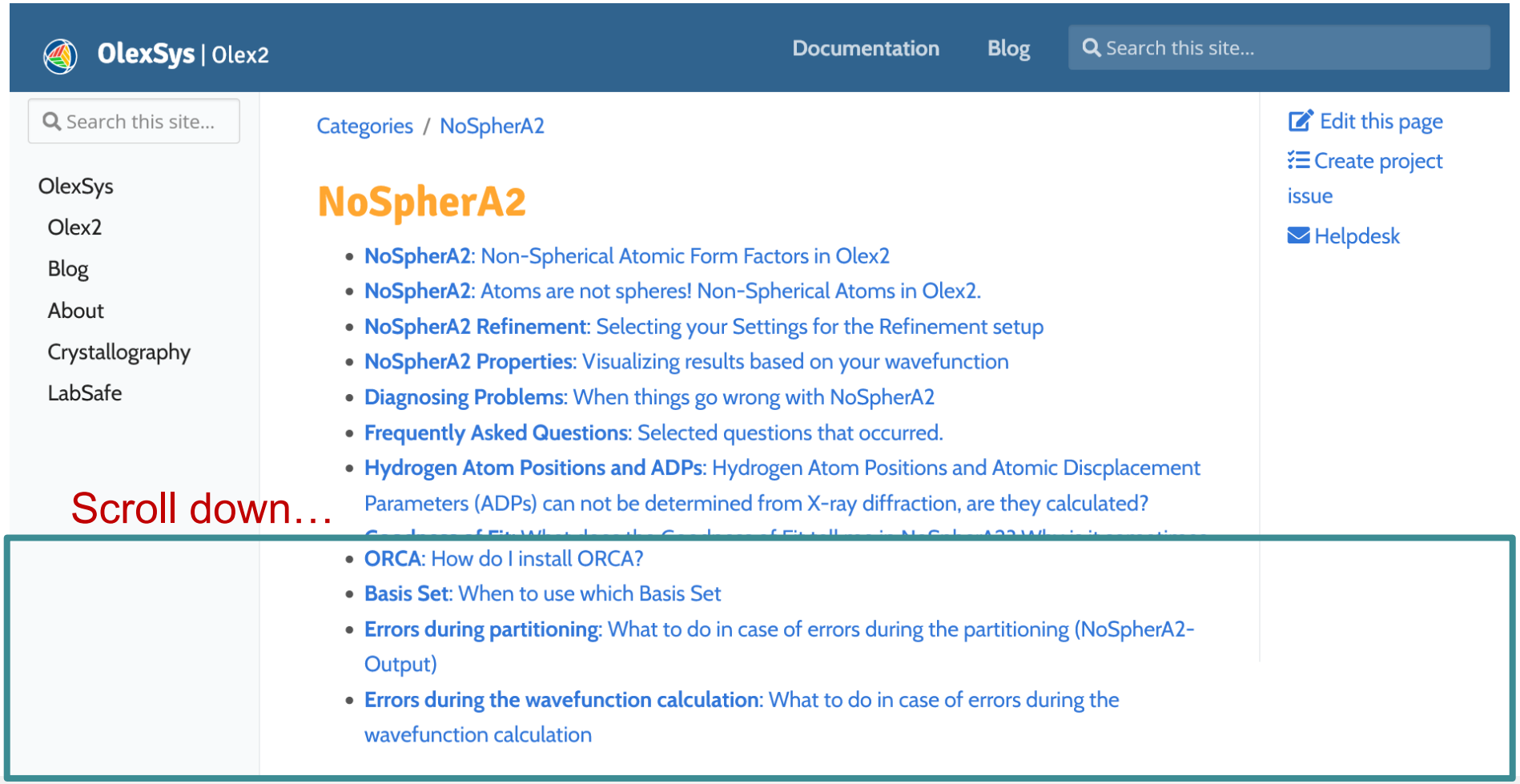
- Obtain Olex2 1.5: https://www.olexsys.org/



- There are two versions: Olex2 1.5 (*standard version*) which is more stable and updates less often.
- Olex2 1.5-alpha – a developer version that is more frequently updated with the latest tweaks and will be what I use in the demos.
- The differences are not usually very great and either standard or developer versions should accomplish everything in this Tutorial.

Installing ORCA

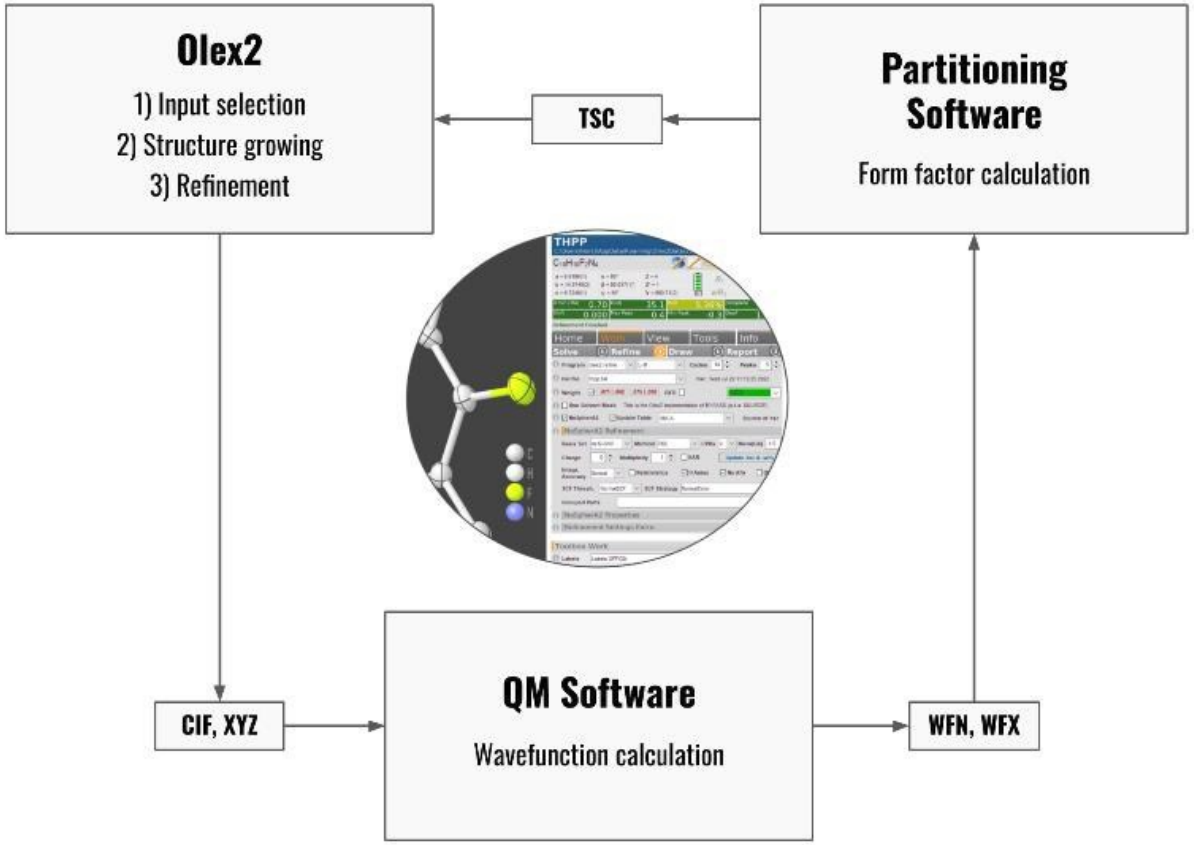
- There are instructions in the Program for this (multi-hour 3 or 4 part download)
- For help and further guidance, see: https://www.olexsys.org/categories/nosphera2/



The currently recommended version of ORCA is 6.1. I have installed 6.1.0, but later minor updates are now(6.1.1) available on the ORCA website.

NoSpherA2 in Olex2

- Flowchart of the process behind HAR controlled by NoSpherA2 and a visual representation of the GUI for NoSpherA2 inside Olex2.



- The starting point is a IAM refinement using olex2.refine
- Use multicycles (30) and update the weights
- Next, implement NoSpherA2 and choose either Test or Work
- Then again implement Refine
- This initiates the ORCA calculation (QM Software)
- When this completes, it opens NoSpherA2 (Partitioning software) to calculate the Aspherical Form Factors (tscb file).
- Then a refinement starts that uses the new AFF to refine the model.
- The process is cyclical and repeated to convergence.

- Florian Kleemis *NoSpherA2: Novel Charge Density Tools in Olex2*
- Watch: https://www.rigaku.com/webinars/single-crystal/no-sphera2-in-olex-2

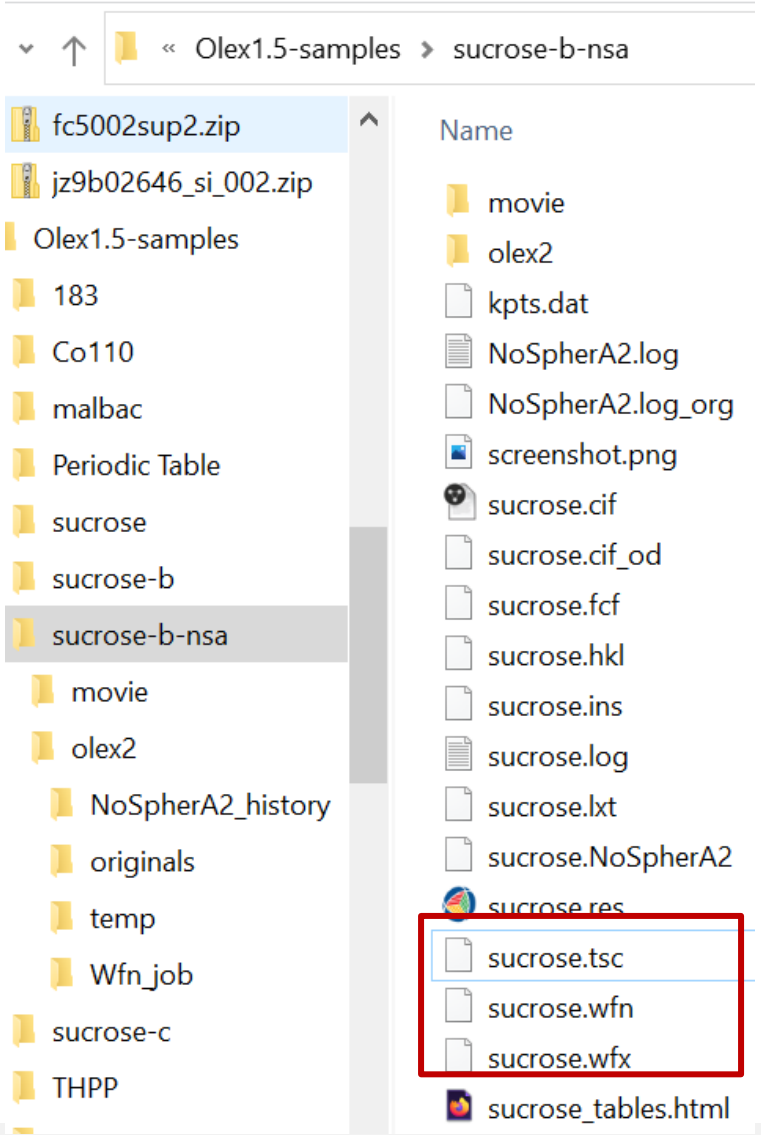
florian@olexsys.org

Features of NoSpherA2

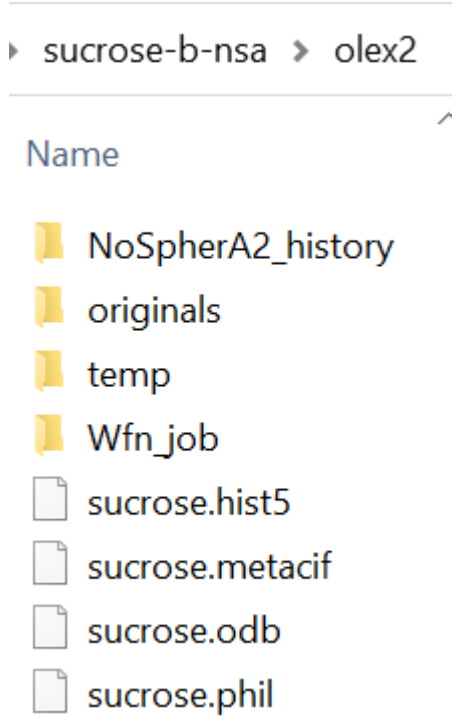
- It works with ‘ordinary’ – albeit good quality – crystallographic datasets
- It works with disorder models – kiss of death for multipole methods!
- It works with open-shell wavefunctions and *e* spin multiplicity (i.e. *paramagnetic* transition metals, lanthanides and MG free radicals)
- It handles heavy elements very well – including relativistic wavefunctions where required
- It works for *any element* and does not use ‘empirical’ atom definitions as in database methods (such as the ‘Invariom’ approach)
- Best of all, it is *fully supported* within Olex2 *and completely controlled by the GUI*
- It is implemented with TONTO, Gaussian for Windows and ORCA – works best with ORCA
- *All the methods for standard crystallographic refinements* are fully supported and enabled
  - Isotropic and anisotropic atom refinements
  - Interatomic interactions – short contacts and hydrogen bonding
  - Full support for symmetry, including atoms on ‘special positions’
  - Use of constraints and restraints in refinement
  - RIGU is particularly helpful for anisotropic H-atom refinement
  - Use of PART commands and partial occupancies that can be refined
  - Use of solvent masks (BYPASS or ‘squeeze’)
- Based on rapid DFT methods easily implemented on a desktop PC!
- It is implemented in the excellent olex2.refine full-featured model refinement method
- Full QM analysis of the outputs can be visualized within the Olex2 GUI

The File Structure in NoSphereA2

The three key files are added to the main structure folder



The olex2 'support folder' grows in size significantly to accommodate the significant DFT and NoSpherA2 operations and store the results.



Why must I use olex2.refine?

- Olex2.refine is written in C++ and Python code – not much older Fortran77, which is used exclusively in SHELXL
- It has an open data structure that allows alternative data sources to be added seamlessly
- The definition files for the form factors easily allow for substitution by alternates

TITLE:	**optional title of the structure**
SYMM:	`expanded' or list of symmetries
SCATTERERS:	space-separated list of all atoms
[ANYTHING]:	colon must be present
DATA:	(denotes the end of the header)

h	k	l	A_1	A_2	…	A_n
h_1	k_2	l_1	$f_1(h_1,k_1,l_1)$	$f_2(h_1,k_1,l_1)$	…	$f_n(h_1,k_1,l_1)$
h_2	k_2	l_2	$f_1(h_2,k_2,l_2)$	$f_2(h_2,k_2,l_2)$	…	$f_n(h_2,k_2,l_2)$
⋮	⋮	⋮	⋮	⋮	⋮	⋮
h_n	k_n	l_n	$f_1(h_n,k_n,l_n)$	$f_2(h_n,k_n,l_n)$	…	$f_n(h_n,k_n,l_n)$

Midgley, L., Bourhis, L. J., Dolomanov, O., Peyerimhoff, N., Puschmann, H. Crystallographic Refinement using Non-Spherical Form Factors in Olex2. refine. arXiv preprint arXiv:1911.08847, **2019**.

```
TITLE: epoxide.cif
SYMM: 1 0 0 0 1 0 0 0 1;-1 0 0 0 1 0 0 0 -1
SCATTERERS: O1 C2 H2a H2b C3 H3a H3b
DATA:
-9 0 1 1.42558118e+00,-7.39703734e-03 1.11903575e+00,3.00122359e-04 8.72665
-9 0 3 1.43530211e+00,-5.21987789e-03 1.12718348e+00,-5.18793337e-03 9.9320
-9 0 5 1.41414548e+00,-3.00424795e-03 1.09412822e+00,-7.48879988e-03 7.7967
-9 1 0 1.40928713e+00,-6.79622200e-03 1.10384187e+00,5.07665937e-04 7.54873
-9 1 1 1.42383652e+00,-6.48364645e-03 1.12085059e+00,-1.00041152e-03 8.8466
-9 1 2 1.43161099e+00,-5.11468321e-03 1.12489786e+00,-6.33712547e-03 9.3687
-9 1 3 1.43149427e+00,-5.11255049e-03 1.12050436e+00,-4.70365583e-03 9.4510
-9 1 4 1.42485345e+00,-4.75333257e-03 1.10994897e+00,-4.53835622e-03 8.6950
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Midgley, L., Bourhis, L. J., Dolomanov, O., Peyerimhoff, N., Puschmann, H. Crystallographic Refinement using Non-Spherical Form Factors in Olex2. refine. arXiv preprint arXiv:1911.08847, **2019**.

"olex2.refine, a powerful code that is comparable to, and compatible with, the trusted, reliable ShelXL method but implemented in modern programming languages in an open-source, extensible environment"

M.A. Ibrahim & R.T. Boeré, New J. Chem. **2022**, *46*, 5479.

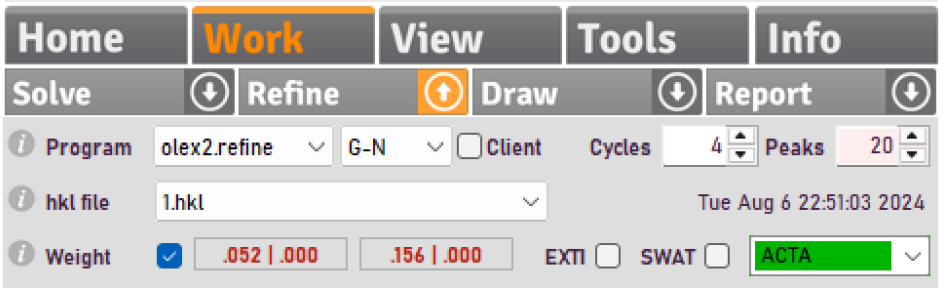
6. More Tutorial Examples

University of Lethbridge NoSpherA2 Workflow (updated)

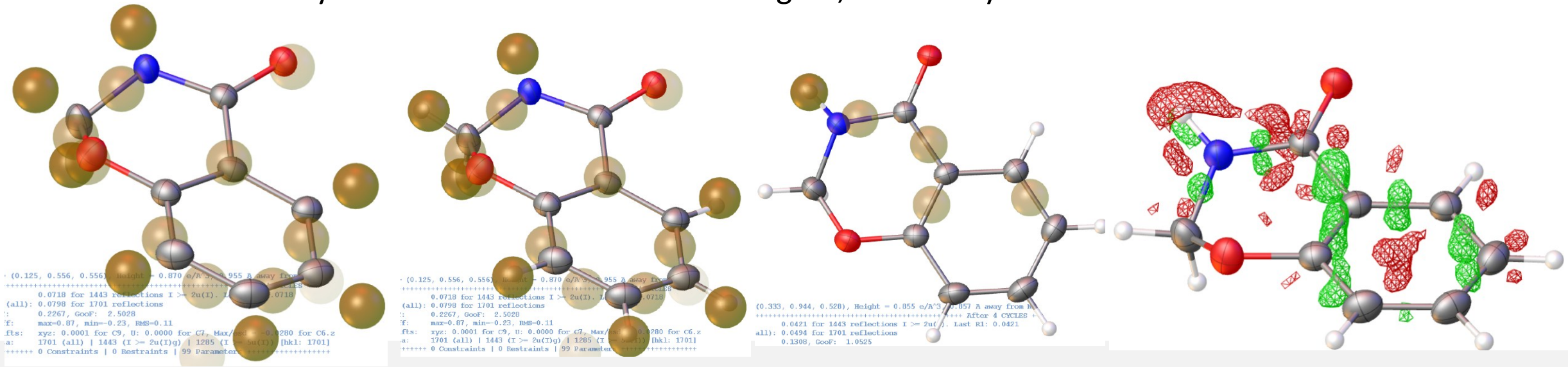
1. Complete the *best possible* refinement model in the IAM (and run in olex2.refine; G-N level; 30 cycles), inc. disorder
2. Carefully check the correctness of the atom labels as this is much harder to change in NoSpherA2
3. *Implement NoSperA2 in a copy of the original folder* (very easy to do in Olex2)
   (i) saves the IAM refinement for later comparison; (ii) very easy to move back and forth
4. Implement the NoSperA2 option (*only available in the olex2.refine page*) and select the default method, or choose a method of your preference – note that NoSpherA2 *automatically* expands E-H bonds to Neutron lengths
5. Obey all the laws of quantum mechanics: avoid 'half bonds' by using GROW for site symmetry; select correct spin multiplicity (hi spin vs. lo spin for metals and biradicals); aggregates like H-bonded dimers can be computed too
6. Step through the 3-levels in NoSpherA2 refinement: Test – Work – Final
   Test: purpose is to test for glitches (naming errors, correct spin state, etc)
   Work: here is where the HAR model is fully developed
   After successful .tscb calculations, refine the XRD model to convergence and adjust weights
   Relax AFIX by selecting H atoms (all or some), then issue **afix 0** at the command line; refine to conv.
   Relax H atoms to ADPs (all, or some by selecting them), then issuing **anis –h** at command line; refine to conv.
   Examine model carefully (badly behaved H may need **rigu** for E-H pairs and/or **isor** restraints)
   Wonky H-geometries (e.g. non-trigonal CH_3 groups) should be reverted to AFIX with Neutron H-distances
   Final: iterative (usually 10x; sometimes more cycles required) for final polishing of the HAR/NoSpherA2 model
7. Validate your work carefully; submit to PLATON CheckCIF; carefully check the final geometry and ensure chemical reasonableness of the model; do a statistical analysis

Demonstrate with Dataset 3167108 – mol1

- Copy the folder from DataCollection-2026 to a working area on your computer
- Open the .ins file with Olex2-1.5alpha
- Type: **kill $h** <retn> to remove all H atoms
- Refine in olex2.refine with ~20 peaks – H-atom peaks obvious!
- Now type: **hadd** <retn> and look for H atoms at calc. positions



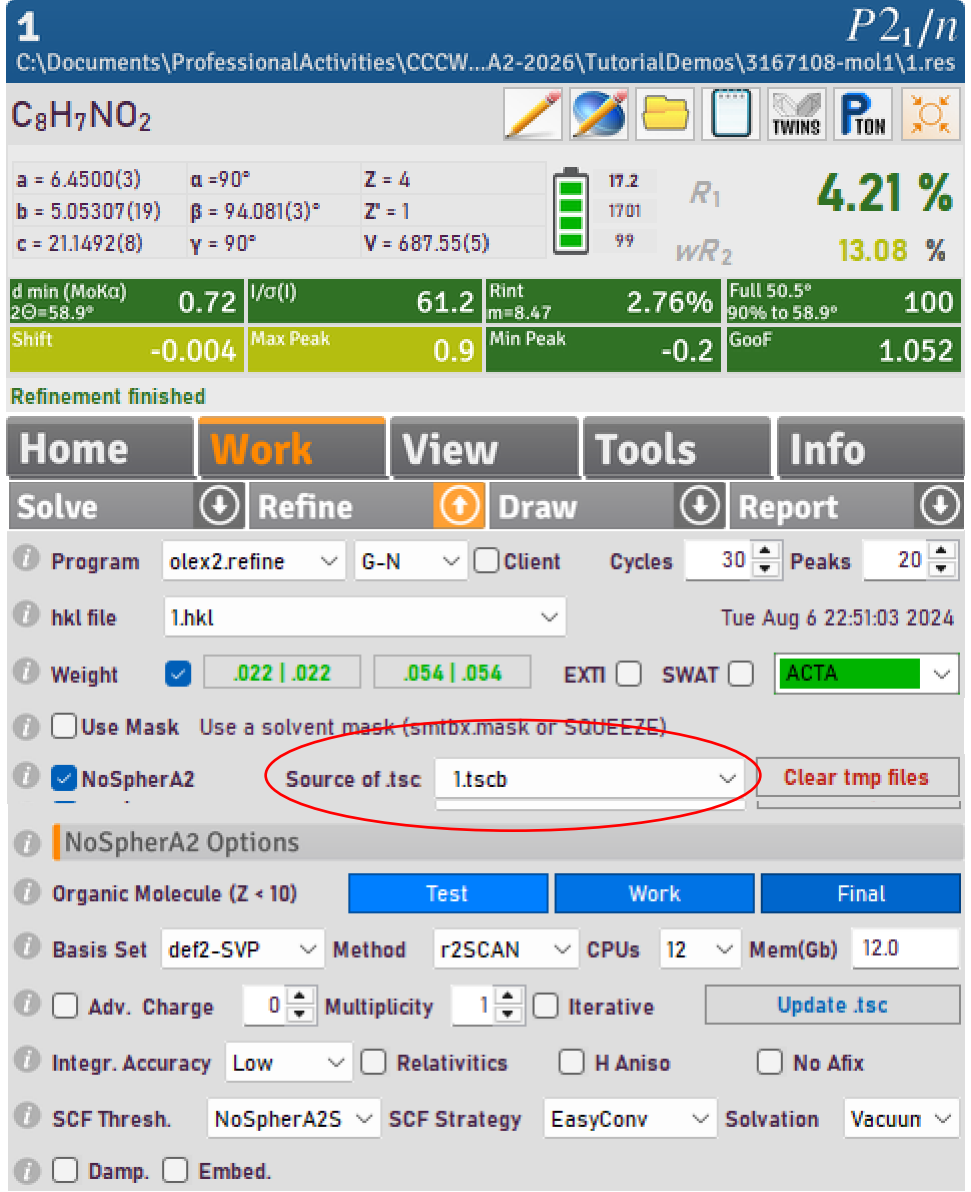
- Olex2 doesn't put an H on O4 (but O2-C4 is 1.237 Å, so a C=O). But N3 does need H! Use Tools – H atoms.
- The planar N requires HFIX 43 (designed for aromatic geometries); 'CTRL+R' to refine.
- 'CTRL+M' already shows clear evidence of bonding ED, noticeably for the benzene π-bonds.



Dataset 3167108, continued

- Refine the IAM model to convergence (5 peaks, adj. weights)
- Go to Report and collect the IUCr CheckCIF report.
- Now use to copy the folder to '-nsa', then Open .res in new folder, then initiate NoSpherA2; note blank 'Source .tsc'
- Now choose ORCA 6.1, and select Work (because I know it!)
- Initiate NoSpherA2, with Refine (or 'CTRL+R')
- After the program runs, a first cycle of olex2.refinement follows
- Now adjust the weights until convergence is re-established

Note 1.tscb!



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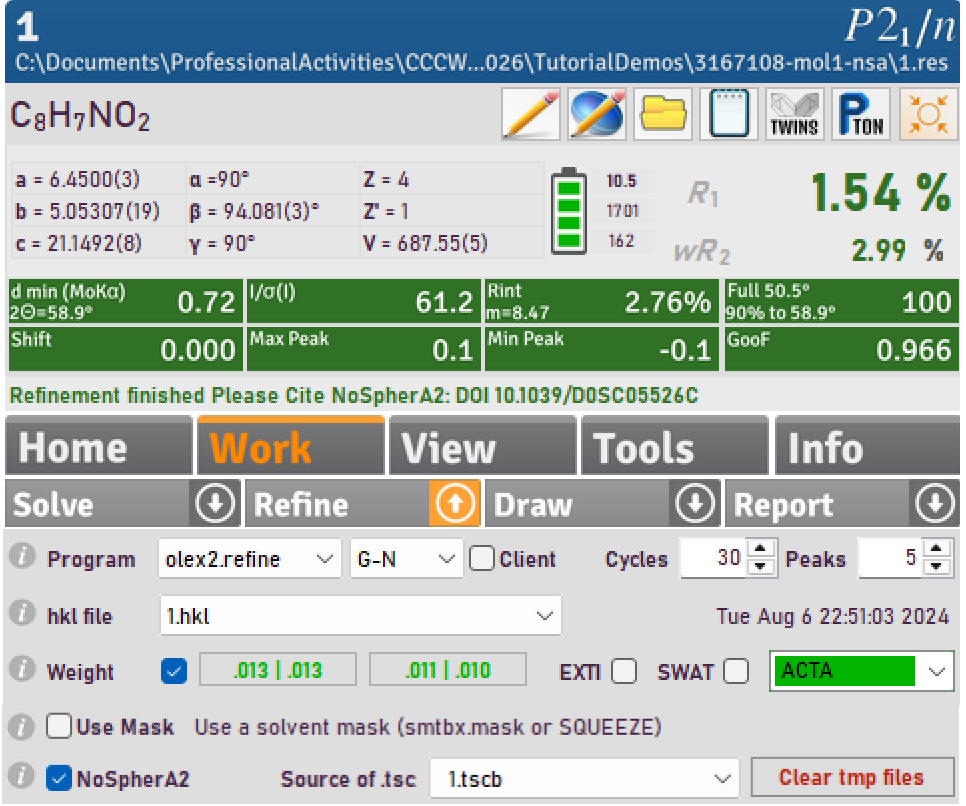
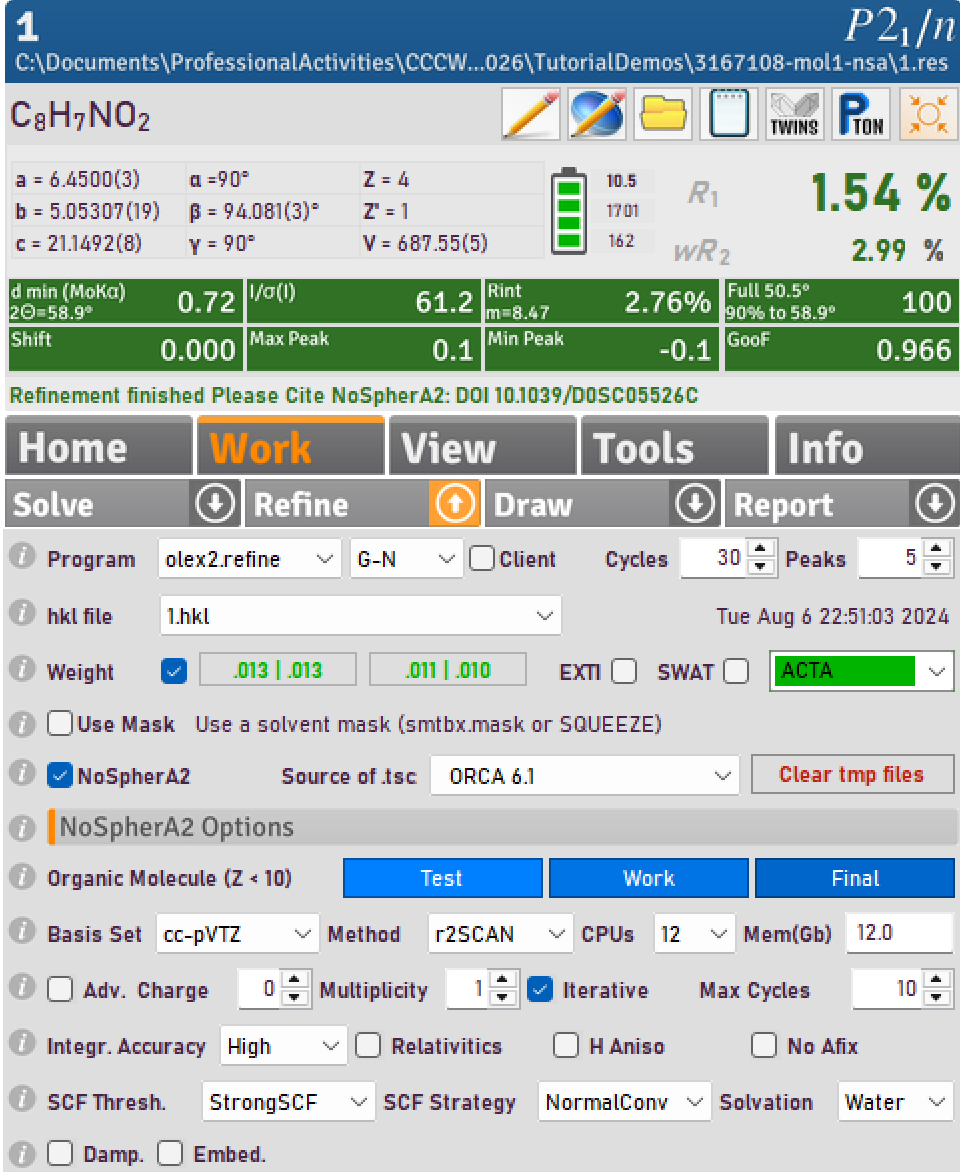
- Now adjust the weights until convergence is re-established

  .179 | .025 .647 | .049 .022 | .022 .054 | .054 Note 1.tscb!
- **sel $h** <retn>; **affix 0** <retn>; CTRL+R; then check the outcome
- **anis –h** <retn>; CTRL+R; *carefully* inspect the outcome
- H7 looks a bit odd, so click on C7 & H7, **rigu** <retn>; CTRL+R
- Looks good all round! Now inspect all C-H an N-H bonds for reasonableness and regularity.



Dataset 3167108, HAR finalization

- With the model in good shape, we finalize the HAR and .tscb calculations, by again choosing ORCA 6.1 as the Source.
- It is now a 10X iterative DFT calculation (warning: long!), with adjustments to Integr. Accur., SCF strategy, Solvation: *water*



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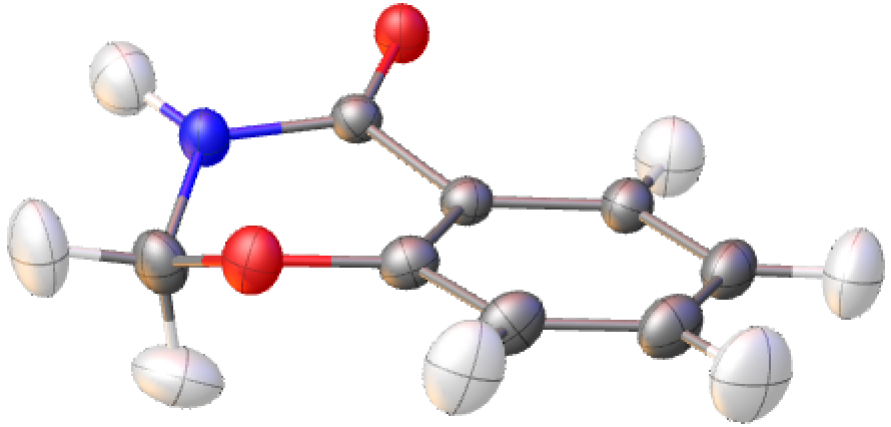
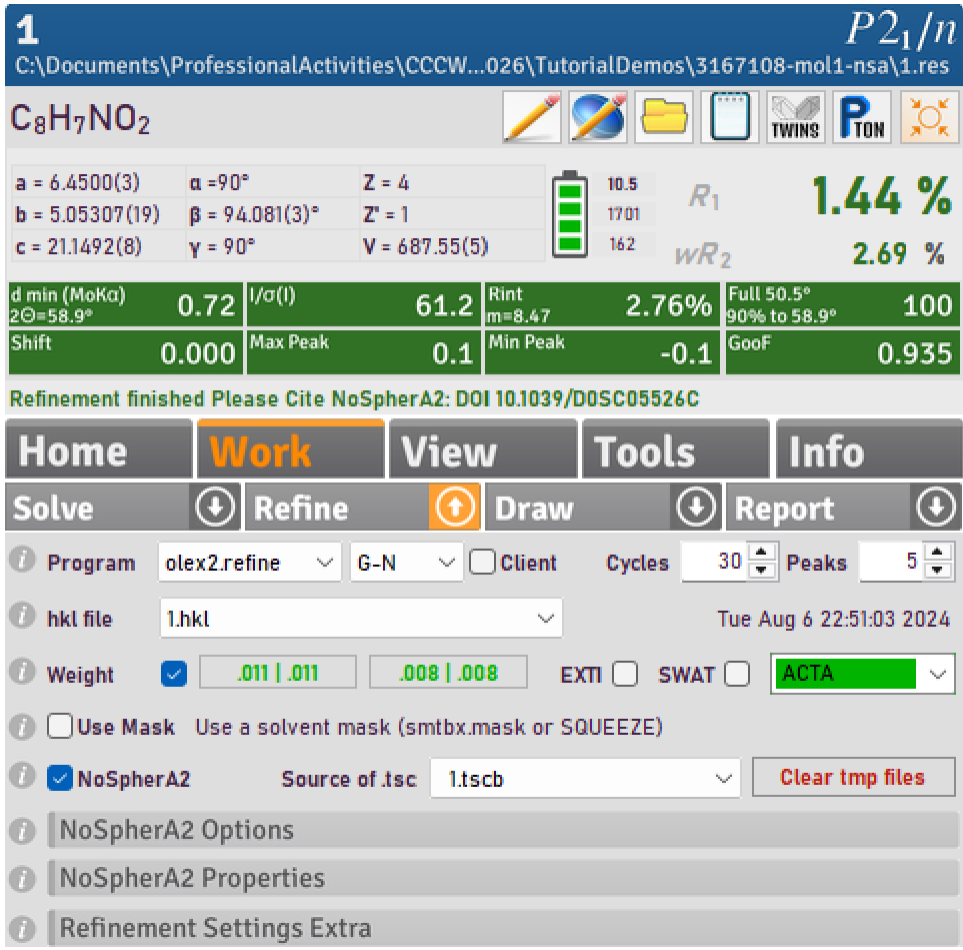
- It is now a 10X iterative DFT calculation (warning: long!), with adjustments to Integr. Accur., SCF strategy, Solvation: *water*
- Hit Refine, and wait. When done, refine olex2.refine to conv.
- Ca. 2 min on my laptop; final R_1 = 1.44%
- Go to Report and collect the IUCr CheckCIF report.
- Bond precision: C-C = 0.0005 Å vs. 0.0016 Å, a 69% improvement





Dataset 3167108, Statistical Analysis on Bond Lengths

Structure 3167108-mol1-nsa

Table 4 Bond Lengths for 1.

Atom1	Atom2	Length/Å	Value	s.u.	Type	Neutr	s.d.	Z-score	Interp.
C2	H2a	1.102(5)	1.102	0.005	CH2(sp3)	1.097	0.006	0.64	same
C2	H2b	1.090(5)	1.090	0.005	CH2(sp3)	1.097	0.006	0.90	same
N3	H3	1.002(5)	1.002	0.005	NH(sp3)	1.019	0.013	1.22	probl same
C6	H6	1.089(4)	1.089	0.004	CH(Ar)	1.085	0.009	0.41	same
C7	H7	1.085(5)	1.085	0.005	CH(Ar)	1.085	0.009	0.00	same
C8	H8	1.088(4)	1.088	0.004	CH(Ar)	1.085	0.009	0.30	same
C9	H9	1.083(5)	1.083	0.005	CH(Ar)	1.085	0.009	0.19	same
C4	C5	1.4810(5)	1.481	0.0005					
C5	C10	1.4010(5)	1.401	0.0005					
C5	C6	1.3963(5)	1.396	0.0005					
C6	C7	1.3861(5)	1.386	0.0005					
C7	C8	1.3965(6)	1.397	0.0006					
C8	C9	1.3885(6)	1.389	0.0006					
C9	C10	1.3908(5)	1.391	0.0005					
		Average s.u. C–C		0.000529	Platon Bond precision: C-C = 0.0006 A				
N3	C2	1.4386(5)	1.439	0.0005					
N3	C4	1.3441(5)	1.344	0.0005					
O1	C10	1.3656(4)	1.366	0.0004					
O1	C2	1.4225(5)	1.423	0.0005					
O4	C4	1.2371(4)	1.237	0.0004					

Interpretation Rules of Thumb:

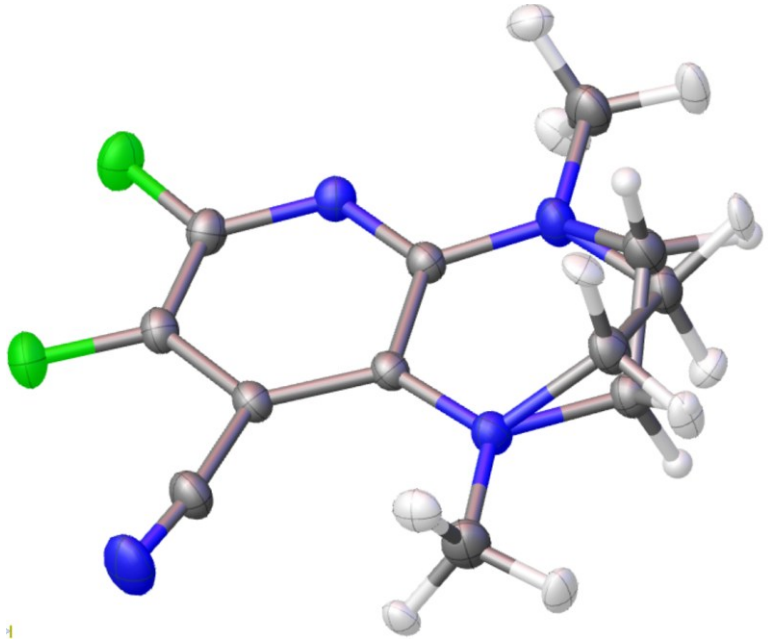
- $Z < 1$: The values are very close; the difference is likely due to noise.
- $Z \approx 2$: There is a 95% chance the difference is real (statistically significant).
- $Z > 3$: The difference is highly significant. Khan Academy +4

7. Resources and Support

A two-part disorder model you can try on Olex2 THPP

THPP: substituted tetrahydropyrido[2,3-b]pyrazine – like sucrose, an Olex2 provided dataset

- There is a conformational disorder in the $-CH_2CH_2-$ component of the hydrated pyrazine
- Minor component is 11.8% and the major 88.2%
- Even with this disorder the major and minor components can be anisotropically refined for non-hydrogen atoms, and for the H atoms of the major component.
- The low occupancy second component H atoms are isotropically refined *and retained in HFIX.*
- The HFIX C–H distance is, in the final tweaking, adjusted to an average NoSpherA2 length:
  AFIX 23 1.09
- IAM: R_1 = 3.93; NOTE: R_{int} = 5.36%
  Max pk 0.3; min pk -0.3
  C-C = 0.0012 Å
- NoSphereA2: R_1 = 2.19;
  Max pk 0.2; min pk -0.2
  C-C = 0.0007 Å



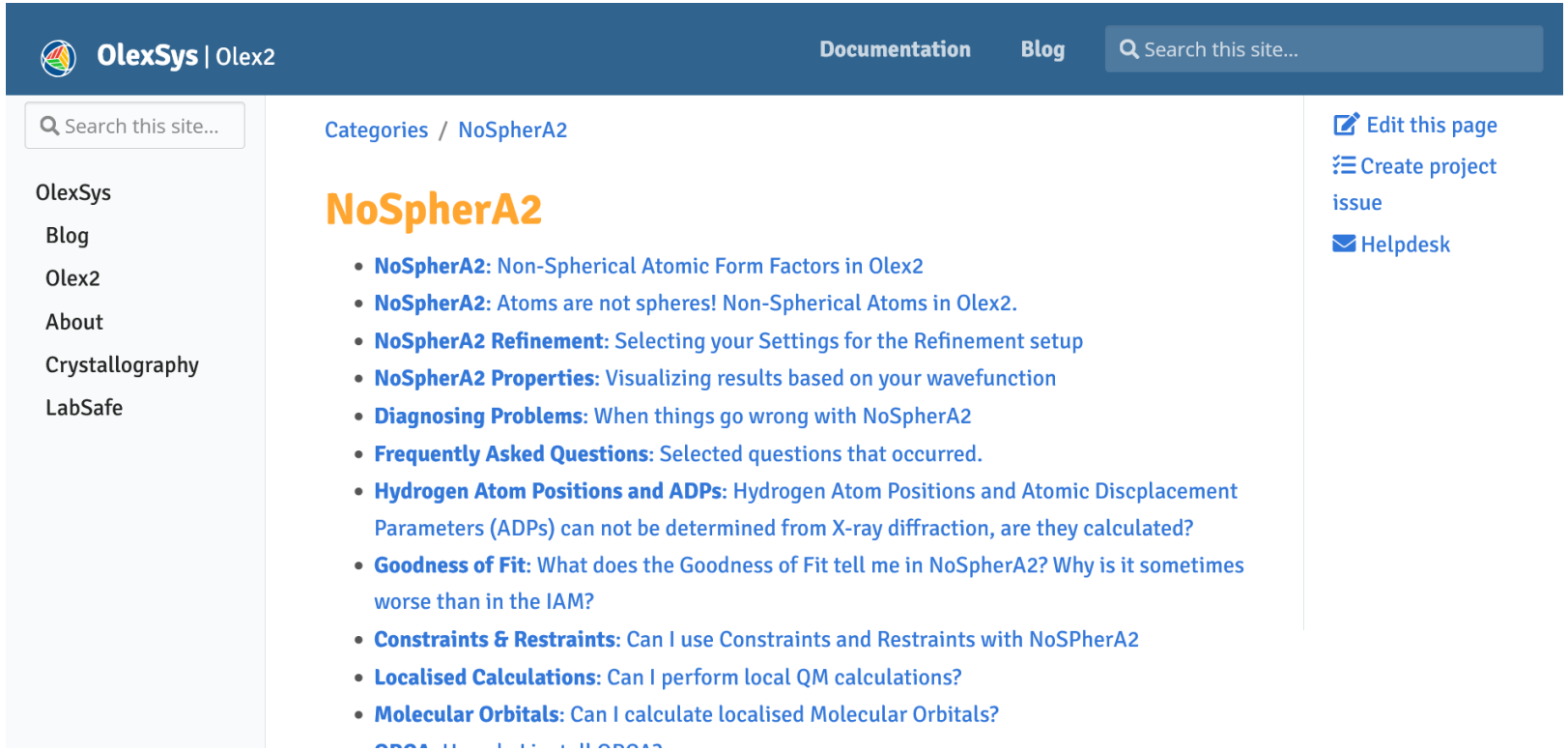
So even with a marginal R_{int} > 5% we obtain about 42% improvement in mean precision.
HAR with NoSpherA2 yields more accurate crystal structures!

Datasets Provided For Further Practice – *or try your own!*

- Via the CCCW website; refinements verified by **Roberto Diaz-Rodriguez** and assigned a difficulty code
- LERFET01: **Easy.**
- 3167108 mol 1: **Easy.**
- 3167108 mol 2: **Easy.**
- 3223140: **Easyish.**
- 3422588: **Hard.**
- 3426677: **Very Hard.** *Caution*: Olex2 sometimes places H atoms that are not present e.g. at N atoms.
- 3429186: **Hard** – *involves a disorder model***.**
- 3642310: **Medium with a caveat.**
- 4017045: **Medium.**
- 4056700: *hard; there are two CH_3 groups on sp^2 C that are rotationally disordered, AFIX 124*
- C13H9N-BenzOctatriene: *easy*
- Ph2-3,5ditBuOHP=O: *medium due to being a rather large molecule*
- TBCP6CN: *easy but note that the GOF will stay 'red'*
- THPP (included within Olex2): *challenging due to the disorder model*

Help from Olexsys.org…and acknowledging their work

- First and foremost, the www.olexsys.org support pages



- Web docs and video links: https://www.olexsys.org/olex2/docs/nosphera2/
- Olex2 has its own YouTube channel!

"If you use NoSpherA2 you agree to cite the following literature in your main manuscript:"
Kleemiss et al., Chem. Sci., 2021, https://pubs.rsc.org/en/content/articlepdf/2021/sc/d0sc05526c

Further information

1. Capelli, S. C.; Burgi, H. B.; Dittrich, B.; Grabowsky, S.; Jayatilaka, D., Hirshfeld atom refinement. *IUCrJ* **2014,** *1* (Pt 5), 361-379.
2. Woińska, M.; Grabowsky, S.; Dominiak, P. M.; Woźniak, K.; Jayatilaka, D., Hydrogen atoms can be located accurately and precisely by x-ray crystallography. *Science Advances* **2016,** *2* (5), e1600192.
3. F. Kleemiss et al. Accurate crystal structures and chemical properties from NoSpherA2, *Chem. Sci.* (2021), **12**, 1675–1692.
4. S. Grabowsky, A. Genoni and H-B. Bürgi, Quantum Crystallography, *Chem. Sci.* (2017), **8**, 4159.
5. F. H. Allen, I. J. Bruno, Bond lengths in organic and metal-organic compounds revisited: X-H bond lengths from neutron diffraction data. *Acta. Crystallogr. (*2010) **B66**, 380-6.

https://www.youtube.com/watch?v=fidBw8sI1Wk

A detailed video lecture that covers much of the ground of this tutorial

Includes a detailed description of the Ph2-3,5ditBuOHP=O dataset.

Available for reminders and/or further details.